Assignment on pset3 and 5 roll 708

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Consider "diamonds" data set in R. It is in the ggplot2 package. Make a list of all the ordinal categorical variables. Identify the response

```
library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.5.2

data=diamonds
head(data)

## # A tibble: 6 × 10
##   carat cut       color clarity depth table price     x     y     z
##   <dbl> <ord>     <ord> <ord>   <dbl> <dbl> <int> <dbl> <dbl> <dbl>
## 1  0.23 Ideal     E     SI2      61.5    55   326  3.95  3.98  2.43
## 2  0.21 Premium   E     SI1      59.8    61   326  3.89  3.84  2.31
## 3  0.23 Good      E     VS1      56.9    65   327  4.05  4.07  2.31
## 4  0.29 Premium   I     VS2      62.4    58   334  4.2   4.23  2.63
## 5  0.31 Good      J     SI2      63.3    58   335  4.34  4.35  2.75
## 6  0.24 Very Good J     VVS2     62.8    57   336  3.94  3.96  2.48
```

The ordinal variables are cut . The response here is the price

(a) Run a linear regression of the response on the quality of cut. Write the fitted regression model.

```
data$cut=factor(data$cut,ordered=FALSE)
data$cut=relevel(data$cut,ref="Ideal")
model=lm(data$price~data$cut)
library(stargazer)

## Warning: package 'stargazer' was built under R version 4.5.2

##
## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary
Statistics Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

stargazer(model,type='text',out="txt")

##
## ===============================================
##                         Dependent variable:
```

```
##                      ---------------------------
##                                  price
## ----------------------------------------------
## cutFair                      901.216***
##                               (102.412)
##
## cutGood                      471.322***
##                               (62.703)
##
## cutVery Good                 524.218***
##                               (45.050)
##
## cutPremium                  1,126.716***
##                               (43.225)
##
## Constant                    3,457.542***
##                               (27.001)
##
## ----------------------------------------------
## Observations                   53,940
## R2                              0.013
## Adjusted R2                     0.013
## Residual Std. Error   3,963.847 (df = 53935)
## F Statistic         175.689*** (df = 4; 53935)
## ==============================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

(b) Test whether the expected price of diamond with premium cut is significantly different from that of the ideal cut.

Now E(Y|X=ideal)=μ_i

E(Y|X=premium)=μ_p

We are to test for $H_0 : \mu_p$ - μ_i = β_p =0 vs $H_1 : \beta_p \neq 0$

Now since there are are three stars beside the coefficient of premium , we reject the null hypothesis at 1 percent level of significance . Hence the average price of diamond with premium cut is significantly different from that of the ideal cut

(c) What is the expected price of a diamond of ideal cut?

E(Y|X=ideal)= β_0 . Here β_0 = 3457.542

(d) Modify the regression model in (a) by incorporating the predictor "table". Write the fitted regression model.

```
model=lm(data$price~data$cut+data$table)
model
```

```
##
## Call:
## lm(formula = data$price ~ data$cut + data$table)
##
## Coefficients:
##       (Intercept)       data$cutFair       data$cutGood  data$cutVery Good
##          -6563.67             345.61             -19.96             165.21
##   data$cutPremium         data$table
##            626.22             179.10
```

The fitted regression model is :

$\hat{Y}$ = -6563.67+345.61 * cutFair - 19.96 * cutVeryGood + 626.22 * cutPremium + 179.10 * table

(e) Test for the significance of "table" in predicting the price of diamond.

```
summary(model)
```

```
##
## Call:
## lm(formula = data$price ~ data$cut + data$table)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -5630  -2694  -1458   1346  15690
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -6563.672    517.450 -12.685  < 2e-16 ***
## data$cutFair        345.611    106.002   3.260 0.001113 **
## data$cutGood        -19.957     67.426  -0.296 0.767246
## data$cutVery Good   165.206     48.562   3.402 0.000669 ***
## data$cutPremium     626.220     50.215  12.471  < 2e-16 ***
## data$table          179.105      9.236  19.393  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3950 on 53934 degrees of freedom
## Multiple R-squared:  0.0197, Adjusted R-squared:  0.01961
## F-statistic: 216.7 on 5 and 53934 DF,  p-value: < 2.2e-16
```

Since there are three stars beside the coefficient of table , We reject the null hypothesis at 1 percent level of significance . Hence it has a significant impact in determining the prices of diamond

(f) Find the average estimated price of a diamond with an average table value and which is of fair cut.

Y=E(Y|x1=table ,x2=faircut)= β_0 + β_1 * faircut + β_2 * table

$\hat{Y}$ = -6563.672 + 345.611 * faircut +179.105 * table

Consider a setting with n = 1000 observations. Generate (i) x1i from N(0, 2^2) and x2i from Poisson(λ = 1.5). (ii) εi from N(0, 1). (iii) yi = −2 + 1.4x1i − 2.6x2i + εi

```
set.seed(123)
x1i=rnorm(1000,0,2)
x1i

##    [1] -1.120951293 -0.460354979  3.117416628  0.141016783  0.258575470
##    [6]  3.430129974  0.921832412 -2.530122469 -1.373705704 -0.891323940
##   [11]  2.448163595  0.719627654  0.801542901  0.221365432 -1.111682270
##   [16]  3.573826274  0.995700956 -3.933234313  1.402711803 -0.945582815
##   [21] -2.135647412 -0.435949829 -2.052008897 -1.457782459 -1.250078536
##   [26] -3.373386621  1.675574089  0.306746236 -2.276273874  2.507629842
##   [31]  0.852928443 -0.590142966  1.790251322  1.756266975  1.643162163
##   [36]  1.377280508  1.107835307 -0.123823421 -0.611925327 -0.760942002
##   [41] -1.389413958 -0.415834556 -2.530792703  4.337911931  2.415923997
##   [46] -2.246217166 -0.805769671 -0.933310707  1.559930237 -0.166738133
##   [51]  0.506637028 -0.057093511 -0.085740915  2.737204568 -0.451541971
##   [56]  3.032941209 -3.097505608  1.169227499  0.247708488  0.431883137
##   [61]  0.759278966 -1.004646906 -0.666414767 -2.037150766 -2.143582453
##   [66]  0.607057283  0.896419557  0.106008453  1.844534936  4.100169371
##   [71] -0.982062332 -4.618337751  2.011477049 -1.418401525 -1.376017233
##   [76]  2.051142739 -0.569546014 -2.441435425  0.362606959 -0.277782725
##   [81]  0.011528372  0.770560802 -0.741320064  1.288753097 -0.440973124
##   [86]  0.663563928  2.193678026  0.870362982 -0.651863171  2.297615237
##   [91]  1.987007712  1.096793919  0.477463470 -1.255812152  2.721304897
##   [96] -1.200519174  4.374665986  3.065221252 -0.471400718 -2.052841801
##  [101] -1.420813127  0.513767418 -0.493383757 -0.695085199 -1.903237135
##  [106] -0.090055450 -1.569808939 -3.335883873 -0.760453041  1.837993218
##  [111] -1.150693925  1.215928644 -3.235765417 -0.111123931  1.038814408
##  [116]  0.602306724  0.211352388 -1.281412017 -1.699408692 -2.048257581
##  [121]  0.235293194 -1.894949228 -0.981114887 -0.512184384  3.687724010
##  [126] -1.303899803  0.470773145  0.155921699 -1.923713268 -0.142616172
##  [131]  2.889101717  0.903008106  0.082465844 -0.844993665 -4.106494443
##  [136]  2.262674427 -2.921280142  1.479895022  3.818207138 -2.887786322
##  [141]  1.403568671 -0.524394979 -3.144288318 -3.029335308 -3.203072347
##  [146] -1.061813044 -2.923511170  1.375833546  4.200217881 -2.574060952
##  [151]  1.575477695  1.538084482  0.664405158 -2.016753217 -0.238905213
##  [156] -0.560790670  1.125979066 -0.744877512  1.953946773 -0.749161716
##  [161]  2.105422931 -2.098354013 -2.520310490  6.482079870 -0.833715176
##  [166]  0.596455183  1.273139348 -0.967561251  1.033724089  0.737929055
##  [171] -0.430761015  0.130586067 -0.068134507  4.256903798 -1.482672193
##  [176] -2.191992534  0.075576798  0.620961499  0.873046958 -0.916730665
##  [181] -2.126652268  2.526370352 -0.699300776 -1.731025725 -0.472559138
##  [186] -0.394351789  2.219840579  0.169474584  1.508107570 -0.998584034
##  [191]  0.428890619 -0.649371823  0.189167056 -1.790726716 -2.621603067
##  [196]  3.994426769  1.201417647 -2.502542723 -1.222331833 -2.370960169
##  [201]  4.397620698  2.624825953 -0.530290113  1.086388118 -0.828679896
```

```
##  [206] -0.952493789 -1.577205676 -1.189234535  3.301814935 -0.108056250
##  [211]  0.238490473  0.487374859  2.464951757 -1.032127662 -1.985014301
##  [216]  3.351393865 -0.882326434 -1.446131940 -2.472546238 -2.569431445
##  [221] -1.147946959  1.235971634  2.219696278  1.415176708 -0.727314594
##  [226]  0.119499875 -1.409192927 -1.434436323  1.769300998 -2.031185157
##  [231]  3.910587931 -0.180639188  0.429077653 -1.477055409 -1.148777380
##  [236] -2.634032265 -0.365850777  0.837964810  0.648608688 -1.563072974
##  [241] -1.577243942 -1.004397437  2.992121340 -2.274607241 -0.358103189
##  [246]  3.804723643 -0.201949771 -2.719681408 -1.329538871  0.970919958
##  [251] -0.751205743 -1.123752727 -0.687834468  0.180993294  3.197017542
##  [256] -0.177130224  2.161598992  1.261508231 -0.227279791 -3.065804006
##  [261] -1.042234635 -0.979740906  0.094308866  2.600397355  4.586157948
##  [266]  3.095162118 -0.266301929 -3.513054791 -0.777559728  0.178414446
##  [271]  1.690026008  1.925055937  1.368618859 -2.790548700  1.699286091
##  [276] -0.893114433  0.349605400  0.149102354  0.856333530  0.049349966
##  [281] -3.334950195  1.472991930  0.772053137 -0.531303251  0.236289022
##  [286]  0.268077291  0.442038937  3.281692332 -0.438100758  0.336130768
##  [291]  2.336767746  2.108362047  2.290526221 -1.154936002  4.004965461
##  [296]  0.133401742  3.733703689 -2.701805372  0.041967173  2.499829142
##  [301] -1.430484374 -1.505377936 -1.877077407 -2.105026559 -0.874319066
##  [306]  0.662358346 -4.028420996  0.423960867  2.473350093  4.075148036
##  [311]  2.602351984  1.513549528 -3.453460798 -1.203013416 -0.704092913
##  [316]  1.407047806 -0.211342668 -2.517297256  3.368871416  1.822782584
##  [321]  0.474860545  2.436217221 -2.677548574  1.321640596 -1.045824753
##  [326]  1.367491044 -0.121643909  1.265921426  2.671035230  0.014580181
##  [331]  2.035117274 -2.376868070 -1.443208881  3.038435423  0.754775946
##  [336] -4.104445641 -2.728074904 -0.401562031  1.731558809 -0.203766511
##  [341]  1.248374944  1.918010756  3.342109658  0.112033467 -0.103963812
##  [346] -3.506474718  0.198655188 -1.143700116 -1.948019166 -0.359812462
##  [351]  2.029886345 -3.985496977 -0.854558574  0.233274567 -1.786415140
##  [356]  0.667805885  0.822859841 -0.066072319 -4.931796388  5.142916292
##  [361] -0.410598515  1.302386563  0.547532982  2.049346470  1.635318893
##  [366] -0.419586342  0.756335544 -1.890817662  1.713846022 -0.922076678
##  [371]  4.833546708 -3.302097791 -0.927974486  1.650759726  1.020265094
##  [376] -1.178962077 -1.993561484  0.288951409 -0.028614826 -3.580562475
##  [381]  0.069102134  0.380460631  0.349452794 -2.110034085  0.952266557
##  [386]  2.757140274  0.912472806 -2.271176941 -0.871290939  0.692207239
##  [391] -1.294091263 -4.315292670  1.768501640 -1.658955223 -1.147120542
##  [396]  3.007801218 -1.548289859  1.691463080 -2.521365758 -0.709084806
##  [401] -0.147112038 -2.337302849 -1.269496530 -0.057683106  1.341391937
##  [406] -3.301093087 -0.699508478  1.512812877 -1.077618320  0.454583843
##  [411]  0.984457140  0.535670031  1.306515359 -0.245417322 -0.827353027
##  [416] -5.286297904 -0.185882037  0.860569393  1.070797682 -1.110556703
##  [421]  3.559005820  0.572848839  0.252631717  2.544533559 -1.436932443
##  [426] -0.900677248  4.794904960  0.022258374  3.267136843 -2.877013290
##  [431] -0.381033604  0.756847807  0.600077090 -2.011272519  0.038518549
##  [436] -2.154841306  1.425406650  2.169550180 -4.449975393  2.471386925
##  [441] -2.482088994  0.909538538  1.319805276 -0.399779656 -1.290227914
##  [446]  0.330642042  0.877637401  1.766605640 -4.104673968 -3.272758536
##  [451]  2.860804682  2.093257694  0.870577898  1.430356814  1.834349835
```

```
##  [456] -5.321845597  2.220554193 -0.969975193  0.461233661 -0.590315602
##  [461]  1.743929908 -0.696944898  1.037007532 -0.781369957 -2.185574418
##  [466]  2.420021021  1.481800023  3.448524478  0.130307865  2.250005492
##  [471]  3.950838108 -0.562964230 -2.645902226 -0.478703134 -0.428082480
##  [476]  0.303361009  3.424609955 -0.652287785  0.746009312 -0.455368130
##  [481]  0.040901417  0.628115327  2.656429392  0.242636755  1.425684640
##  [486]  1.557720060  1.829546542 -1.148789104  3.253762428 -0.761913479
##  [491] -0.211568335  2.808100535  2.588167812 -2.179983745 -1.746142001
##  [496] -2.716158119  0.363694385  0.329681736  0.728229375  1.104315428
##  [501] -1.203785692 -1.987397182  2.053570112  1.502122605 -3.018333075
##  [506] -0.190294902 -1.791895646 -4.141502141  0.300240262 -0.158423419
##  [511] -0.194738535  0.432305084  1.764930329  0.411195009 -1.232871686
##  [516] -1.469598504 -0.263605586  0.620033973 -2.079360705 -0.368617738
##  [521]  1.934534520 -0.216560182 -1.396841335 -0.551890337  2.229297091
##  [526]  1.100087922  2.473351600  0.278195714  0.820550193 -1.116913825
##  [531]  1.210741338 -1.012667084 -2.841131009  0.255985931  3.891702435
##  [536]  1.601828679  2.330506780  0.717711446 -1.217114357 -0.404481710
##  [541] -0.546496214 -0.937399560  1.408334568 -2.394727005  1.732732264
##  [546]  1.728304972 -2.397244718  1.278983996  4.860453305 -1.114430964
##  [551]  1.689808483 -1.564403691  2.221422836  0.499649440  3.303830783
##  [556] -2.917941457 -0.102595772 -1.053850352 -0.394529738 -1.259157485
##  [561] -1.667687161  1.157444749 -2.175161428  2.968061863 -2.372413171
##  [566]  0.202158303  1.065978574  1.173470678 -0.603493328  0.159003991
##  [571]  1.922528304 -2.912931834 -1.563479423  0.640804629 -0.889563957
##  [576]  2.740007988  1.346507727  0.144333505 -3.015514635  0.052200451
##  [581] -0.632831736 -0.204693027 -2.363118454  0.997316088 -2.077912881
##  [586] -0.452443963  0.762851659 -1.567031578  1.165982811 -2.633020803
##  [591] -5.619549358  0.929935982  1.681079654 -0.571690847  1.008252509
##  [596] -2.311833050 -0.254297214 -3.883036752  2.362361783  3.719821723
##  [601]  2.148024521 -0.054693933 -0.066660685 -3.032135246  1.580770688
##  [606] -0.421468361 -1.313485851 -2.824051585 -0.599524993 -1.698122270
##  [611] -0.794061041 -2.435199976  3.375178966 -0.032005054  2.149890155
##  [616] -5.203399341 -0.906395660 -1.350964583 -2.445852359  3.093218308
##  [621] -2.830563840  0.636780521  1.692872571  0.356380377 -1.750510955
##  [626]  1.882331613  0.341176164 -2.126995813 -2.776098095  4.173434869
##  [631] -1.357006291 -3.711143308  1.066518726  0.620460512 -2.707668671
##  [636] -3.885912827 -0.232605034  2.278792583  1.272248070 -0.985874848
##  [641] -1.668376463  0.542133514  0.314706694  1.259423493 -0.791595901
##  [646]  1.798708106 -1.661623055 -0.661089400  1.481629047  1.979943217
##  [651] -3.877009403  0.214380814  1.217558023 -2.901648627  0.961251200
##  [656] -1.656348541  2.040506024  1.076964064  1.538104590  0.241438659
##  [661]  1.727296868  2.761029057  3.932496031 -0.056790109 -4.498102181
##  [666]  0.063052009  0.411122417 -0.310690694  1.136577242  2.021355929
##  [671] -1.035964860 -0.588190654  0.795684418 -1.100447483  0.182534767
##  [676] -3.923415198 -2.239799443 -2.655510951 -1.707247393 -1.386609056
##  [681]  0.764610285  1.964226005 -1.454767058 -1.993677965 -2.083377718
##  [686] -0.829177453 -0.478058136  0.967235068 -0.642649684 -4.156978538
##  [691] -0.182868551  2.374373621  2.383202537 -1.577926436 -3.095553087
##  [696]  4.916120984 -0.324843885 -0.194902500  0.841148377 -3.228078913
##  [701] -1.456438222 -3.080884811 -1.386189228  0.237698866 -2.729418916
```

```
##  [706]  1.179965359  0.578688058 -1.808430051  0.452649885  1.496162325
##  [711]  2.122190507 -0.425696557 -0.187273588 -0.173428270  2.882923512
##  [716]  2.250143785  1.668803136 -0.574681600  0.746482868  0.806580662
##  [721] -2.083346589 -3.456609031  1.283660055 -3.058621062  0.003367377
##  [726]  0.500495643  1.127734781  0.378852475 -1.465707613  1.972731720
##  [731]  3.477267534  1.762357618 -3.887301803  2.799152371 -0.112111892
##  [736]  1.049828559  1.244066472 -0.193372146 -0.150526395  2.038314139
##  [741]  1.423203845  1.980524492  4.765853390  1.328831727  0.414762315
##  [746] -4.421266221  5.383428006 -0.965353644  4.749469430  0.749287136
##  [751]  3.076860398 -0.219420642  1.022941509  0.427915959 -0.372241399
##  [756] -0.240787649  2.025668672 -0.402916294 -4.075364989 -0.391778498
##  [761]  1.079581211  1.232911433  1.233135635 -3.384203042  0.737484116
##  [766]  1.935718420  2.553157363 -0.449922542 -0.643785172  2.975675664
##  [771] -3.335856092 -0.873659954  0.914924159 -3.235547531  0.559255725
##  [776]  3.755728042 -0.008121307 -0.556908050  0.949823428 -0.558144342
##  [781]  1.626800748  1.808870929  0.005383322 -2.353384315 -2.636441455
##  [786] -1.185994731  1.594761001 -3.916410349 -3.772650317 -1.307559650
##  [791]  0.788789696 -1.827132096  1.773498074  0.666739940 -0.341279236
##  [796]  1.637656273  0.776730326 -0.891870054  0.462229869  1.295026716
##  [801]  0.712566690 -1.316020416  1.710404410  2.305872451  0.552549113
##  [806]  0.288209325 -0.151250162  4.322831693  0.552631064 -0.316588057
##  [811] -5.015835604 -3.130563530 -0.155346401  0.412588090  0.553744918
##  [816]  1.643013561 -0.388304819  2.429177589 -1.843032086 -2.416885441
##  [821] -2.457972355  1.484594048 -0.165839885  1.579635844 -0.535412843
##  [826] -1.183784204 -0.736705150 -3.705233648 -2.339230521 -2.884069293
##  [831]  2.108644544 -1.194660183  1.578919706  3.032981203 -0.383549618
##  [836]  0.567757813 -3.502135039 -1.637339555  0.112429699  0.598173807
##  [841] -1.518796235  5.369717998 -0.916780278  0.128487113  1.299583734
##  [846] -0.052037270 -1.287134774  2.090611328  3.231090646 -0.059387942
##  [851]  1.124534691 -0.194824997  2.032910435 -2.312334789  4.641720448
##  [856] -1.207062496 -2.917698829 -0.701835654  0.293416968  3.247242415
##  [861]  1.822419354  0.284916864 -2.778967048 -1.732075489 -0.326569864
##  [866]  5.106052227 -3.720455148  2.262109304 -1.054468514  3.331981800
##  [871] -2.278401280  0.287246469 -2.199101883  1.807032853  2.967558977
##  [876]  3.901442013  1.595201315  3.686532509  2.492847817 -0.263749824
##  [881]  0.954074477 -1.943988423 -0.370404339  2.441927417  1.082568287
##  [886]  0.914714663 -2.076262079 -1.209026461 -1.529212028  0.790591740
##  [891] -1.981015261  1.124082780 -2.232832824  3.657060916  0.921182699
##  [896] -1.402007225  0.482091865 -0.704906398  0.742295920  0.487065438
##  [901] -2.028228345 -1.582627758  0.599187369  3.278103818  2.169234017
##  [906] -1.249134947  1.651845803 -0.097136706  0.602627303  0.520722981
##  [911]  5.150899527 -2.370577621  0.201839718 -3.559954576  1.179671846
##  [916]  2.193216943  2.891324482 -3.850290504  0.825538994  3.186739903
##  [921] -0.828031726 -0.424301064 -0.073074444  0.730037503  1.330319751
##  [926]  2.635641767 -0.190975181  0.392556091  4.975995754  0.862197856
##  [931]  0.377506219 -2.684486251  0.005712095 -0.442652306 -0.022091660
##  [936] -1.150835283 -1.373631305 -1.441547264 -0.429009031  2.736265296
##  [941]  2.098173253 -0.719950236 -3.371832909 -1.689166857 -0.915521066
##  [946]  0.207276008 -1.325214551  4.013361382 -0.544535069 -2.427888940
##  [951] -0.282523515 -2.010755163  0.312311415  0.467267228  0.711175223
```

```
##  [956] -3.243716518  0.441422582  0.620900161 -2.842216895  1.910731279
##  [961]  1.568341758  4.599238722  0.313405975  0.093467055  0.193171668
##  [966]  0.139532462 -3.696945551 -3.342254117 -0.155077934 -1.162134761
##  [971]  0.109473050 -4.222416747 -2.997396509 -2.202966878  1.972116442
##  [976] -2.196980014 -1.599027907  0.159747637 -0.645492724  0.292834357
##  [981]  4.610123964 -2.249207343 -0.610939280 -1.033518899  3.024790854
##  [986] -1.538969846 -0.164173808  1.574267229 -2.117181072  3.310351631
##  [991]  1.351524831 -2.148413219  0.909155618 -0.426614286  0.626457544
##  [996] -0.179950394  2.141032074 -2.702200771 -1.045233394 -0.498381356
```

```
x2i=rpois(1000,1.5)
x2i
```

```
##    [1] 0 0 0 1 1 2 1 0 0 0 3 1 2 2 5 0 2 1 1 0 6 1 6 0 0 1 2 2 1 2 2 2 2 0
0 2 2
##   [38] 4 4 2 1 0 1 2 3 3 3 0 0 1 2 0 4 0 3 0 2 2 2 2 0 3 2 3 0 0 1 1 3 3 0
3 2 1
##   [75] 2 2 1 1 4 0 0 2 2 0 1 2 1 5 2 1 0 1 0 0 1 2 4 1 1 0 2 1 1 4 2 2 0 0
3 3 2
##  [112] 1 0 1 1 0 2 0 2 3 3 2 1 0 3 0 3 1 1 3 0 1 0 0 2 2 1 3 4 1 2 0 1 0 0
0 1 1
##  [149] 2 2 1 4 2 0 1 1 1 3 1 1 4 0 2 2 2 6 0 5 2 1 0 1 0 1 3 0 1 3 0 2 4 2
0 1 2
##  [186] 0 3 2 1 2 2 0 2 3 3 2 2 2 1 0 3 2 2 0 0 2 0 2 2 1 1 2 0 0 1 0 0 2 2
1 1 2
##  [223] 3 0 5 0 0 1 3 0 2 3 1 2 1 1 4 1 0 4 1 2 0 4 3 3 1 2 1 1 7 5 1 2 1 1
2 1 0
##  [260] 5 2 3 1 0 2 0 0 3 0 1 1 1 0 0 3 4 4 1 1 0 1 3 1 1 1 0 1 0 2 1 1 3 0
1 3 1
##  [297] 1 3 2 3 1 3 0 2 2 0 2 2 3 1 1 0 1 2 2 3 0 2 0 2 2 1 1 2 1 1 1 0 1 1
0 0 2
##  [334] 2 2 0 0 4 3 3 2 0 3 3 0 0 2 1 1 1 2 3 1 2 1 0 0 1 0 1 1 2 1 3 1 1 1
2 2 2
##  [371] 3 1 0 1 1 1 3 1 0 4 1 0 4 1 5 0 2 2 0 2 2 0 1 1 3 5 0 1 2 3 2 2 1 3
2 1 1
##  [408] 2 2 1 3 2 0 0 3 0 1 1 2 1 2 3 1 1 2 1 0 3 1 2 2 1 0 2 0 1 0 2 0 2 2
1 0 0
##  [445] 0 3 1 0 1 2 2 1 0 2 3 5 2 1 2 1 0 2 4 0 1 1 0 0 1 0 0 0 1 2 1 1 2 0
3 1 1
##  [482] 1 2 1 3 0 0 2 1 2 2 1 2 3 2 3 3 1 2 0 3 2 2 1 2 1 2 2 3 1 0 1 0 1 2
2 1 2
##  [519] 1 1 2 2 2 0 1 4 1 0 1 4 3 1 2 1 3 0 1 2 3 3 3 0 2 2 2 3 1 4 1 0 2 0
1 1 3
##  [556] 2 1 0 1 1 1 3 1 1 1 3 0 1 0 3 1 1 1 1 0 0 0 0 1 2 2 0 4 0 1 2 3 3 1
1 1 2
##  [593] 2 0 0 3 1 2 3 1 1 2 1 3 0 0 2 0 2 1 2 1 0 1 1 2 0 2 1 4 3 0 1 2 2 1
1 0 1
##  [630] 0 2 2 1 1 0 0 3 2 2 1 1 4 0 0 4 2 7 1 2 0 2 3 3 0 1 2 1 2 0 2 1 1 2
0 1 2
##  [667] 1 1 1 0 0 0 2 0 0 0 1 3 1 1 1 1 3 3 1 2 0 4 0 5 0 1 0 2 2 6 0 3 2 5
```

```
1 2 0
##  [704] 2 0 3 3 3 3 1 4 1 1 1 0 3 0 1 2 2 4 0 1 3 3 3 2 0 0 0 1 2 3 1 3 1 2
2 2 3
##  [741] 1 0 0 0 3 2 0 0 0 1 2 0 0 1 1 1 0 3 2 1 2 1 2 1 2 1 2 1 2 2 0 1 1 1
1 3 2
##  [778] 0 0 1 1 2 0 4 3 2 3 0 2 3 0 1 2 1 3 1 2 2 1 3 0 1 1 0 2 2 1 3 2 4 3
1 2 3
##  [815] 3 0 0 5 1 2 1 1 2 3 0 1 2 0 2 1 2 2 2 3 1 1 3 1 2 3 1 1 3 2 2 2 2 1
1 0 4
##  [852] 1 1 0 2 1 1 3 1 2 2 0 0 1 2 2 2 3 0 1 2 0 0 0 2 3 1 1 1 4 1 2 3 2 1
3 2 0
##  [889] 2 3 0 4 0 0 0 1 2 2 1 0 4 2 2 1 2 2 3 1 1 2 2 1 2 1 3 2 2 2 2 2 0 2
2 1 2
##  [926] 0 2 4 2 2 1 0 1 0 2 1 2 1 0 1 1 3 0 0 0 4 1 2 0 1 0 2 3 0 1 3 0 2 1
0 3 0
##  [963] 2 1 2 0 3 0 0 1 2 1 1 0 1 1 5 0 3 0 2 1 0 0 0 3 5 1 0 2 2 1 3 1 3 0
3 0 1
## [1000] 2
```

```
ei=rnorm(1000,0,1)
ei
```

```
##    [1] -0.820986697 -0.307257233 -0.902098009  0.627068743  1.120355028
##    [6]  2.127213552  0.366114383 -0.874781377  1.024474863  0.904758894
##   [11] -0.238248696 -1.557854904  0.761309895  1.129144396 -0.295107831
##   [16]  0.536242818 -0.275890475  0.682315245 -0.117290715 -0.344675864
##   [21]  0.111620498 -0.283405315 -0.591017164 -0.315936931 -0.008152152
##   [26]  0.207495141  1.532423622 -1.357997831 -0.199619051  0.631523128
##   [31]  1.762020903  0.426014363 -0.013753416 -0.307556910  0.414308164
##   [36]  0.989057920 -0.183858311  0.163761407  0.216936344  0.729277634
##   [41]  1.111380407  0.279160817 -0.076170672  1.394663132  0.164534118
##   [46]  1.577851979 -0.061922658  0.613922964 -1.546088594 -0.112391961
##   [51] -0.021794540 -0.758345417 -1.035892884  0.948159303  0.914158734
##   [56] -1.298731995  0.424378795 -1.112545320 -1.051073226  0.525412448
##   [61] -0.686024000  0.993479982  0.038523599  0.536148976 -0.523626698
##   [66] -1.151221335  0.914752241  0.238071492 -0.239067759  0.069235327
##   [71]  1.325908343 -0.698166635 -0.749408444 -0.619615053 -1.584991268
##   [76]  0.819628138  0.192369647  0.207171974 -0.043347354 -0.510160441
##   [81] -0.823418614  0.851856403 -1.426184673  0.440298942 -0.792611651
##   [86]  0.282310215 -0.740690522 -0.523341683  1.769365917  0.668282619
##   [91] -2.144897024  0.126412416 -0.451812936 -1.136626188  0.209785890
##   [96]  0.129965516 -0.328506573  1.972703567 -2.248690067  0.838219387
##  [101] -0.289023270  0.656513411 -0.453997701 -0.593864562 -1.710379666
##  [106] -0.209448428  2.478745801  0.989702208  1.675572156  0.914965318
##  [111]  1.144262708  0.902876414  0.475392432 -0.582528774 -0.532934737
##  [116] -1.600839996 -0.005817714  0.899355676  1.031922557  0.095132704
##  [121] -0.547627617  3.290517443  0.736685531  1.420575305 -0.337680641
##  [126] -0.037957627  0.448607098  1.676522312 -0.311474545  0.853615667
##  [131] -2.094814634 -0.507254434 -1.292009077  1.113362717 -0.164453088
##  [136] -0.390374082  1.369099846  1.116272858 -0.898021203  0.427866488
```

```
##  [141] -1.228444569 -0.475615024  1.616577637  1.450127951  1.109018755
##  [146] -0.570903886 -1.881431470 -1.175698184  0.952556525 -0.290567886
##  [151] -2.162608146 -0.180187488  1.410239221  0.643468641 -0.821258544
##  [156] -1.545916652 -0.826547226  0.034527671  0.888073701 -1.939940155
##  [161]  1.023201755  0.005457727  0.569778970 -1.653255563 -0.666654380
##  [166] -0.448234189  1.043891348  1.028174047  0.435090459  1.604212182
##  [171] -0.515411200  1.012537194 -0.035940030 -0.667342096  0.923380038
##  [176]  1.381100331  0.878250416 -0.509403455 -0.469787634  1.377675847
##  [181]  0.352826406  0.829573979 -0.338701984  1.261034936 -0.808755145
##  [186]  0.625351521 -0.817174966 -2.462575017 -1.342957511  0.136295199
##  [191]  0.882922750 -1.751302083 -1.251424469  1.764545997 -0.433899350
##  [196]  0.505700132 -0.526935321 -0.298582885  0.087244207  0.010961843
##  [201] -0.192560211 -0.469796485 -3.047860890  1.868655500  1.790424208
##  [206] -1.101081742 -0.168107522  1.375275305  0.998290023  1.276601619
##  [211] -1.071746918  2.577268096 -1.133459960  0.753916342  0.141275980
##  [216] -0.403710318 -0.379415798 -0.991396813  1.622659796  0.089513228
##  [221]  0.259217949  0.209632833 -0.375170753 -1.134021245  0.253726310
##  [226] -2.093639450 -1.418566943 -1.076396695 -1.078678863  0.107188817
##  [231]  1.598487548 -1.515324144  0.433676021  0.899544753 -0.989532202
##  [236] -0.052799402  0.823610903 -0.255509103 -0.220684347  0.307726791
##  [241] -0.060013253 -0.555652890 -0.138615019  1.882839792  0.873668680
##  [246] -0.914597073 -1.244917622 -0.359982241  1.328774701  0.292679118
##  [251] -0.701505237  0.882234568 -0.133370389 -1.120678499  0.461192454
##  [256]  1.524142810  0.434468298  0.192000371 -0.656243128  0.568398531
##  [261] -1.070570535 -1.653149024 -0.043352768 -0.034593506  2.365055532
##  [266] -1.216347308  0.170906323  0.805053094  1.050592844 -0.010724485
##  [271] -0.743256141 -0.065784052  1.939755992  0.482739008 -2.044477073
##  [276]  1.423459129  0.540502661 -0.033571772 -0.017863621 -0.149789720
##  [281]  0.256559481 -0.503866933  0.277011252 -0.931356025  0.200146875
##  [286]  1.106837421  0.509206114  1.033749676 -1.090868762  0.054792784
##  [291]  0.617250303 -1.068004868  1.565814337 -1.034808011  0.164518709
##  [296]  0.151832330  0.121670302 -0.210424584  0.449936787 -1.031164492
##  [301] -1.289364188 -0.654568638 -0.057324104  1.256747820  1.587454140
##  [306]  0.319481463  0.381591623 -0.243644884  0.048053084 -1.404545861
##  [311]  0.289933729 -0.535553582  0.334678773 -0.345981339 -0.661615735
##  [316] -0.219111377 -0.366904911  1.094578208  0.209208082  0.432491426
##  [321] -1.240853586  1.496821710  0.159370441 -0.856281403  0.309046645
##  [326]  0.870434030 -1.383677138  1.690106970 -0.158030705  1.121170781
##  [331]  0.072261319 -0.332422845 -1.834920047 -1.100172219 -0.041340300
##  [336]  0.827852545 -1.881678654  1.375441112  1.398990464 -1.143316256
##  [341]  0.472300562 -1.033639213 -0.125199979  0.928662739  0.868339648
##  [346] -0.849174604 -0.386636454 -0.976163571  0.339543660 -1.559075164
##  [351] -2.629325442  1.469812282  2.273472913 -0.455033540  0.761102487
##  [356] -0.007502784  1.474313800  0.554143933  0.203663965 -1.799136452
##  [361]  1.082955681 -0.350853615 -1.403490085 -0.201796665 -0.126778160
##  [366]  1.059206873 -1.167396032 -0.557643627  1.488119928  1.358665769
##  [371]  1.163214544  1.661523945  0.204030980 -0.581883687  0.555204062
##  [376]  1.058723126  2.413633271 -1.964982333  0.273235703  0.654794583
##  [381] -0.054598655 -1.557822248  0.741500892 -0.779085741  0.505861499
##  [386]  0.907551706  1.283957010 -1.557863797  1.081741848 -0.756981357
```

```
##  [391] -1.289019474  1.314320666  1.146259973 -0.242583268  0.759540706
##  [396] -0.860325741 -0.151031579 -0.093723234 -0.280740055  0.734098736
##  [401]  1.537327535 -0.455771063 -0.032658452  1.636757349 -0.329041971
##  [406] -2.604038171  0.513983786 -0.886468011 -0.998538406  1.420816814
##  [411]  2.447998007 -1.039782541  1.031025182 -0.094147838  0.141807459
##  [416]  1.222236701  0.213674518 -0.851365354 -0.470408875  0.686135255
##  [421] -2.335947332  1.095244382 -1.567150099  0.021931063 -0.190358978
##  [426]  1.293069486  0.188849315  0.101939135  0.698135809 -0.827014556
##  [431] -0.195898856  1.177584414  0.683473621 -1.275496707  0.637956367
##  [436] -1.377589625 -0.598310804  1.210920384 -2.251045175 -1.779014191
##  [441]  1.301372673 -0.814792784  1.243707019 -0.168250201  0.427775678
##  [446]  0.813278890 -0.651211869 -0.304590923 -0.415097170  2.816084276
##  [451]  0.126147069  0.472800423 -0.340753541 -0.241790643  1.378754669
##  [456] -0.338883670  0.020136303  0.376962164 -0.431723745  1.959064156
##  [461] -1.428459612  2.011292980 -0.351591889  1.357119647 -1.999177411
##  [466]  0.956080621  0.876431265 -1.271216972 -0.768323878  0.193524845
##  [471]  1.143835429 -0.765999304 -0.224125999  1.571346935 -1.127347241
##  [476]  0.947793981  0.448768189 -1.105814529 -0.667867842  0.783277510
##  [481]  0.248959433  1.425098277 -0.601783962 -1.714487696  1.047826926
##  [486] -0.608621618  0.120340527  1.719041814 -0.250414052  1.549555327
##  [491] -1.097139650  0.925511236  0.246799212 -0.736771539 -1.280008939
##  [496]  0.076643664  0.255164762  0.277446817  0.536856022 -0.460485566
##  [501] -0.511603722  0.236937879 -0.541589172  1.219227647  0.174135875
##  [506] -0.615268316 -1.806892964 -0.643681106  2.046018852 -0.560762421
##  [511] -0.835999306  0.652947503  0.441293125  0.751629057 -0.277975085
##  [516]  1.122654216 -1.172608860 -0.048876768 -0.704140337  0.680758638
##  [521]  0.130006761  1.109708082  2.058500869  0.140655532 -0.534616652
##  [526] -0.823516727 -0.263033982 -0.069601836  1.991801915 -1.129109543
##  [531] -1.093217437 -0.407966690  0.587559463  0.821111860 -0.907934699
##  [536]  0.127038606 -0.042892978  1.195206470  1.089192237 -0.312280694
##  [541]  0.045993767  0.652722613 -1.653492637 -0.310270965  0.574872884
##  [546] -0.523232154 -0.059918201 -0.021007535 -0.723653214 -0.994479836
##  [551] -0.199867235 -0.347027823  0.834095066  1.529882212 -0.011922379
##  [556]  0.398671992 -0.070415314  0.601359844  0.218495462  0.236595502
##  [561]  1.112915128 -0.987421149  1.447864013  0.349112412  0.180822011
##  [566] -0.560241851 -0.163877592  0.373684804 -2.063714257 -0.601521949
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##  [916] -1.255444278 -1.080306847  0.175396079  0.330839221 -0.320689231
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```

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yi=-2+1.4*x1i-2.6*x2i+ei
yi
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##  [506]  -5.48168118 -11.51554687 -13.64178410  -7.33364478  -5.38255521
##  [511]  -3.10863325  -3.34182538   0.91219559  -3.27269793  -9.20399545
##  [516]  -8.13478369  -6.14165668  -6.38082921  -8.21524532  -4.43530619
##  [521]  -4.36164491  -6.39347617  -7.09707700  -2.63199094  -2.01360073
##  [526] -11.68339364  -1.40034174  -1.68012784  -1.45942782 -15.09278890
##  [531]  -9.19817956  -6.42570061 -10.59002395  -3.42050784  -5.25955129
##  [536]   0.36959876  -1.38018349  -4.99999750 -10.41476786 -10.67855509
##  [541] -10.51910093  -2.65963677  -6.88182424 -10.86288877  -4.19930195
##  [546]  -7.90360519  -8.01606081 -10.63042994   1.48098141  -4.55468319
##  [551]  -5.03413536  -4.53719299  -0.65591296  -2.37060857  -5.18655928
##  [556] -10.88644605  -4.81404940  -2.87403065  -4.93384617  -6.12622498
##  [561]  -5.82184690  -9.16699850  -6.19736199  -0.09560098  -7.74055643
##  [566] -10.07722023  -0.67150759  -2.58345625  -4.90860492 -10.17891636
##  [571]  -1.32246876  -8.97258635  -7.58939874  -4.33856805  -3.00964051
##  [576]   0.20117880   0.75634006  -3.96686777  -9.32506001  -7.91410185
##  [581]  -9.33456464  -3.36447758 -15.45828848  -0.72353151  -7.80993067
##  [586] -10.15418532 -10.05632839 -12.12515132  -3.84565922  -9.08299805
##  [591] -11.41782839  -5.72250128  -5.89033311  -3.26906320  -0.87334997
##  [596] -13.71686145  -5.92006971 -12.68805553  -5.75149878   0.83460242
##  [601]  -1.79291268  -6.88875126  -3.89940659 -14.18550330   0.66888416
##  [606]  -3.73562861  -9.28853088  -6.37397049  -7.84367049  -6.62005166
##  [611]  -8.43530344  -8.77549419   1.79553634  -4.36628647  -0.23331693
##  [616] -15.27189467  -3.65375260  -8.76066986  -8.57881375  -7.94792205
```

```
##  [621] -13.81038549  -1.88475886  -1.39853715  -5.85475964  -8.62657583
##  [626]  -0.69673915  -4.62871516  -5.44227603  -8.22531933   4.47288979
##  [631]  -9.43943496 -12.81894544  -3.72514531  -2.24915339  -8.29890249
##  [636]  -7.60785599 -10.08743417  -5.06929999  -5.03342681  -7.94731247
##  [641]  -5.98075819 -13.30437367  -3.76214551  -1.00037094 -13.34615387
##  [646]  -5.33337582 -23.08555927  -5.19232019  -6.18462026   0.68607096
##  [651] -12.12988017  -7.86587720  -7.61596689  -4.34754509  -2.80108829
##  [656]  -9.52212908  -3.99982642  -6.91690886  -0.16561620  -6.14971542
##  [661]  -2.50429796  -0.19091070  -2.75831691  -2.35363587 -10.68033702
##  [666]  -7.47111291  -3.91159692  -5.70571500  -2.63444664   0.74884341
##  [671]  -3.49740015  -4.77225400  -6.75971040  -5.03027056  -3.35016938
##  [676]  -7.98666438  -7.89651759 -13.23411511  -5.89888370  -6.09685238
##  [681]  -2.51747526  -2.37639388 -12.14451405 -11.50598027  -7.51552162
##  [686] -10.04109315  -3.51583691 -10.03827885  -3.51044682 -20.48632582
##  [691]  -2.24179328  -1.77223454   0.98569716  -9.01737646 -11.32419549
##  [696]  -9.48276048  -2.65460122 -10.99607154  -5.85648917 -18.81397593
##  [701]  -4.35704656 -11.97692175  -4.26701849  -5.98472838  -4.53990036
##  [706]  -8.80673035  -8.32526626 -12.89695959 -10.12846843  -1.88201184
##  [711]  -9.32243552  -4.80664430  -5.44268652  -3.04782162   2.70138093
##  [716]  -7.02420113   1.03907332  -6.61906862  -6.29267411  -4.66742917
##  [721] -15.50552454  -5.92876227  -3.02479792 -16.38009589 -10.67549822
##  [726]  -8.87657041  -4.17461860  -2.06300866  -3.77601165  -0.20299488
##  [731]  -0.74828169  -5.51001598 -13.87316046   0.25912341  -9.36329149
##  [736]  -2.01477746  -5.88273194  -6.71114407  -7.20110768  -7.99546113
##  [741]  -3.43857684   0.82278722   4.87782481  -0.46099400 -10.21582400
##  [746] -15.48066465   5.96203469  -3.64676681   5.19841809  -5.09688982
##  [751]  -4.14573138  -2.41852077  -1.98069542  -5.98387151  -4.33754255
##  [756]  -4.03623337  -0.19402750 -10.63614008 -14.03948389  -4.83206298
##  [761]  -5.71826461  -3.74338445  -6.24782765 -10.39996545  -5.73114798
##  [766]  -1.31898536  -3.24910480  -6.07278126  -9.88746887  -2.50317841
##  [771]  -6.84725748  -5.86251629  -2.28697819 -10.02328238  -2.67302666
##  [776]  -4.95517224  -7.92455765  -2.98541741  -0.28022747  -5.58861772
##  [781]  -3.22298617  -4.94920498  -4.53439445 -16.20324972 -13.03505182
##  [786]  -9.02965240  -6.87900688  -6.99699206 -11.83606370 -10.97453856
##  [791]  -2.63427519  -7.15382525  -3.08703536  -4.14704931  -9.82498849
##  [796]  -2.16388749  -5.55556532  -8.72064819  -4.60117748  -7.91500175
##  [801]  -0.79459180  -6.62776087  -2.17399316   1.63957337  -7.20261513
##  [806]  -5.65682929  -2.61098996  -2.27083030  -6.88073436 -14.66611055
##  [811] -16.76797189  -8.10251572  -6.63937826 -10.45212345 -10.13790563
##  [816]   0.43396362  -1.91754540 -10.72621971  -6.36385294 -11.55161511
##  [821]  -9.35376635  -4.53408811  -6.92724314  -6.76039825  -2.41372729
##  [826]  -7.31642233  -6.66367046  -7.55747373  -8.69588437  -8.08629500
##  [831]  -3.05758699  -8.54192203  -5.05416464  -6.56637438  -5.69548366
##  [836]  -3.85224690 -14.42091499  -6.92549459  -7.22057041  -8.77907115
##  [841]  -7.25068677   2.38166774 -12.53919709  -6.17384657  -5.33365040
##  [846]  -7.35647641  -9.74290730  -1.92091800  -1.16325595  -3.13244047
##  [851] -12.74460320  -3.89105622  -1.62796131  -6.35404508   0.46219523
##  [856]  -5.66529581  -7.94239609 -11.00834048  -4.61208825  -2.75191352
##  [861]  -4.24392225  -0.80120178  -4.30108472  -7.53397609  -8.67276256
##  [866]   0.04933178  -9.38653302  -7.06166282  -2.33057470  -0.17832369
```

```
##  [871] -10.86830503  -2.30827207  -5.28998726   2.17163046  -2.74357071
##  [876]  -3.85065206  -1.52798237   2.63288703  -0.33439421 -14.19636111
##  [881]  -4.29780707 -11.51103891 -13.16711284  -2.49056769  -3.57812827
##  [886]  -8.12442879  -8.92514906  -4.20446974  -9.47586449  -8.34291538
##  [891]  -4.54754214 -11.60059936  -4.39515034   3.66552081  -0.99879352
##  [896]  -7.78519103  -5.89173779  -6.75934930  -2.18026822  -0.44547382
##  [901] -15.72265475  -9.94702578  -6.94882244  -0.42235252  -3.45388675
##  [906]  -8.69239217  -9.34377646  -6.59657902  -3.77915587  -6.32104908
##  [911]  -2.29621500  -8.73525590  -6.88986324  -8.12215049 -10.16132814
##  [916]  -5.38494056  -4.23245257 -12.41501063  -5.71340619  -3.05925337
##  [921]  -4.77157310  -8.42457391  -7.86279021  -3.78052878  -3.71466717
##  [926]   1.01312794  -7.39110085 -12.55581981  -1.47383379  -5.35697510
##  [931]  -5.12211997  -3.02307156  -4.49944013  -2.55945965  -7.29747353
##  [936]  -4.36752386  -8.45915672  -6.86915681  -3.76680245  -1.80795635
##  [941]  -2.44754675  -9.59298190  -6.90954765  -5.12203222  -2.48967002
##  [946] -10.76463357  -7.14983186  -2.02622661  -2.41706491  -8.00388895
##  [951]  -1.98916645  -8.30085870  -9.42315057  -1.62652815  -3.11894023
##  [956] -14.39054766  -0.75424332  -6.55471093  -8.13558094  -0.88871692
##  [961]  -7.59041788   3.92271822  -7.95177421  -4.88221533  -6.55756472
##  [966]  -1.71231159 -14.28224001  -5.73825652  -1.38864508  -6.55147784
##  [971]  -8.37489393 -10.79171842  -7.62669968  -5.20562999  -3.47632863
##  [976]  -7.18438880 -16.95681976  -2.17695666 -10.53032831  -0.22064423
##  [981]   0.55336964  -8.20518717  -2.84465013  -4.90101560   1.50695387
##  [986]  -9.94631739 -13.73183365  -2.62528061  -5.65651865  -3.93213101
##  [991]  -3.18181338  -7.49314878  -9.12113104  -4.11919266 -10.02254527
##  [996]  -1.52536635  -5.36168479  -5.99325124  -4.61204581  -7.25618247
```

```
data=data.frame(x1i,x2i,yi)
```

Split the data into train and test sets. Keep the first 800 observations as train- ing data and the remaining as test data. Work out the following:

```
train=data[1:800,]
train
```

```
##              x1i x2i           yi
## 1   -1.120951293   0  -4.39031851
## 2   -0.460354979   0  -2.95175420
## 3    3.117416628   0   1.46228527
## 4    0.141016783   1  -3.77550776
## 5    0.258575470   1  -3.11763931
## 6    3.430129974   2  -0.27060448
## 7    0.921832412   1  -2.94332024
## 8   -2.530122469   0  -6.41695283
## 9   -1.373705704   0  -2.89871312
## 10  -0.891323940   0  -2.34309462
## 11   2.448163595   3  -6.61081966
## 12   0.719627654   1  -5.15037619
## 13   0.801542901   2  -5.31653004
## 14   0.221365432   2  -5.76094400
## 15  -1.111682270   5 -16.85146301
```

```
## 16   3.573826274   0   3.53959960
## 17   0.995700956   2  -6.08190914
## 18  -3.933234313   1  -9.42421279
## 19   1.402711803   1  -2.75349419
## 20  -0.945582815   0  -3.66849181
## 21  -2.135647412   6 -20.47828588
## 22  -0.435949829   1  -5.49373508
## 23  -2.052008897   6 -21.06382962
## 24  -1.457782459   0  -4.35683237
## 25  -1.250078536   0  -3.75826210
## 26  -3.373386621   1  -9.11524613
## 27   1.675574089   2  -3.32177265
## 28   0.306746236   2  -8.12855310
## 29  -2.276273874   1  -7.98640247
## 30   2.507629842   2  -3.05779509
## 31   0.852928443   2  -4.24387928
## 32  -0.590142966   2  -7.60018579
## 33   1.790251322   2  -4.70740157
## 34   1.756266975   0   0.15121686
## 35   1.643162163   0   0.71473519
## 36   1.377280508   2  -4.28274937
## 37   1.107835307   2  -5.83288888
## 38  -0.123823421   4 -12.40959138
## 39  -0.611925327   4 -13.03975911
## 40  -0.760942002   2  -7.53604117
## 41  -1.389413958   1  -5.43379913
## 42  -0.415834556   0  -2.30300756
## 43  -2.530792703   1  -8.21928046
## 44   4.337911931   2   0.26773984
## 45   2.415923997   3  -6.25317229
## 46  -2.246217166   3 -11.36685205
## 47  -0.805769671   3 -10.99000020
## 48  -0.933310707   0  -2.69271203
## 49   1.559930237   0  -1.36218626
## 50  -0.166738133   1  -4.94582535
## 51   0.506637028   2  -6.51250270
## 52  -0.057093511   0  -2.83827633
## 53  -0.085740915   4 -13.55593016
## 54   2.737204568   0   2.78024570
## 55  -0.451541971   3  -9.51800003
## 56   3.032941209   0   0.94738570
## 57  -3.097505608   2 -11.11212906
## 58   1.169227499   2  -6.67562682
## 59   0.247708488   2  -7.90428134
## 60   0.431883137   2  -6.06995116
## 61   0.759278966   0  -1.62303345
## 62  -1.004646906   3 -10.21302569
## 63  -0.666414767   2  -8.09445708
## 64  -2.037150766   3 -12.11586210
## 65  -2.143582453   0  -5.52464213
```

```
## 66   0.607057283   0  -2.30134114
## 67   0.896419557   1  -2.43026038
## 68   0.106008453   1  -4.21351667
## 69   1.844534936   3  -7.45671885
## 70   4.100169371   3  -3.99052755
## 71  -0.982062332   0  -2.04897892
## 72  -4.618337751   3 -16.96383949
## 73   2.011477049   2  -5.13334058
## 74  -1.418401525   1  -7.20537719
## 75  -1.376017233   2 -10.71141539
## 76   2.051142739   2  -3.50877203
## 77  -0.569546014   1  -5.20499477
## 78  -2.441435425   1  -7.81083762
## 79   0.362606959   4 -11.93569761
## 80  -0.277782725   0  -2.89905626
## 81   0.011528372   0  -2.80727889
## 82   0.770560802   2  -5.26935847
## 83  -0.741320064   2  -9.66403276
## 84   1.288753097   0   0.24455328
## 85  -0.440973124   1  -6.00997402
## 86   0.663563928   2  -5.98870029
## 87   2.193678026   1  -2.26954129
## 88   0.870362982   5 -14.30483351
## 89  -0.651863171   2  -6.34324252
## 90   2.297615237   1  -0.71505605
## 91   1.987007712   0  -1.36308623
## 92   1.096793919   1  -2.93807610
## 93   0.477463470   0  -1.78336408
## 94  -1.255812152   0  -4.89476320
## 95   2.721304897   1  -0.58038725
## 96  -1.200519174   2  -8.75076133
## 97   4.374665986   4  -6.60397419
## 98   3.065221252   1   1.66401332
## 99  -0.471400718   1  -7.50865107
## 100 -2.052841801   0  -4.03575913
## 101 -1.420813127   2  -9.47816165
## 102  0.513767418   1  -3.22421220
## 103 -0.493383757   1  -5.74473496
## 104 -0.695085199   4 -13.96698384
## 105 -1.903237135   2 -11.57491165
## 106 -0.090055450   2  -7.53552606
## 107 -1.569808939   0  -1.71898671
## 108 -3.335883873   0  -5.68053521
## 109 -0.760453041   3  -9.18906210
## 110  1.837993218   3  -6.31184418
## 111 -1.150693925   2  -7.66670879
## 112  1.215928644   1  -1.99482348
## 113 -3.235765417   0  -6.05467915
## 114 -0.111123931   1  -5.33810228
## 115  1.038814408   1  -3.67859457
```

```
## 116  0.602306724   0  -2.75761058
## 117  0.211352388   2  -6.90992437
## 118 -1.281412017   0  -2.89462115
## 119 -1.699408692   2  -8.54724961
## 120 -2.048257581   3 -12.57242791
## 121  0.235293194   3 -10.01821715
## 122 -1.894949228   2  -6.56241148
## 123 -0.981114887   1  -5.23687531
## 124 -0.512184384   0  -1.29648283
## 125  3.687724010   3  -4.97486703
## 126 -1.303899803   0  -3.86341735
## 127  0.470773145   3  -8.69231050
## 128  0.155921699   1  -2.70518731
## 129 -1.923713268   1  -7.60467312
## 130 -0.142616172   3  -9.14604697
## 131  2.889101717   0  -0.05007223
## 132  0.903008106   1  -3.84304309
## 133  0.082465844   0  -3.17655690
## 134 -0.844993665   0  -2.06962841
## 135 -4.106494443   2 -13.11354531
## 136  2.262674427   2  -4.42262988
## 137 -2.921280142   1  -7.32069235
## 138  1.479895022   3  -6.61187411
## 139  3.818207138   4  -7.95253121
## 140 -2.887786322   1  -8.21503436
## 141  1.403568671   2  -6.46344843
## 142 -0.524394979   0  -3.20976799
## 143 -3.144288318   1  -7.38542601
## 144 -3.029335308   0  -4.79094148
## 145 -3.203072347   0  -5.37528253
## 146 -1.061813044   0  -4.05744215
## 147 -2.923511170   1 -10.57434711
## 148  1.375833546   1  -3.84953122
## 149  4.200217881   2  -0.36713844
## 150 -2.574060952   2 -11.09425322
## 151  1.575477695   1  -4.55693937
## 152  1.538084482   4 -10.42686921
## 153  0.664405158   2  -4.85959356
## 154 -2.016753217   0  -4.17998586
## 155 -0.238905213   1  -5.75572584
## 156 -0.560790670   1  -6.93102359
## 157  1.125979066   1  -3.85017653
## 158 -0.744877512   3 -10.80830085
## 159  1.953946773   1  -0.97640082
## 160 -0.749161716   1  -7.58876656
## 161  2.105422931   4  -8.42920614
## 162 -2.098354013   0  -4.93223789
## 163 -2.520310490   2 -10.15865572
## 164  6.482079870   2   0.22165625
## 165 -0.833715176   2  -9.03385563
```

```
## 166  0.596455183   6 -17.21319693
## 167  1.273139348   0   0.82628644
## 168 -0.967561251   5 -15.32641170
## 169  1.033724089   2  -5.31769582
## 170  0.737929055   1  -1.96268714
## 171 -0.430761015   0  -3.11847662
## 172  0.130586067   1  -3.40464231
## 173 -0.068134507   0  -2.13132834
## 174  4.256903798   1   0.69232322
## 175 -1.482672193   3 -10.95236103
## 176 -2.191992534   0  -3.68768922
## 177  0.075576798   1  -3.61594207
## 178  0.620961499   3  -9.44005736
## 179  0.873046958   0  -1.24752189
## 180 -0.916730665   2  -7.10574708
## 181 -2.126652268   4 -15.02448677
## 182  2.526370352   2  -2.83350753
## 183 -0.699300776   0  -3.31772307
## 184 -1.731025725   1  -5.76240108
## 185 -0.472559138   2  -8.67033794
## 186 -0.394351789   0  -1.92674098
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## 188  0.169474584   2  -9.42531060
## 189  1.508107570   1  -3.83160691
## 190 -0.998584034   2  -8.46172245
## 191  0.428890619   2  -5.71663038
## 192 -0.649371823   0  -4.66042264
## 193  0.189167056   2  -8.18659059
## 194 -1.790726716   3 -10.54247141
## 195 -2.621603067   3 -13.90414364
## 196  3.994426769   2  -1.10210239
## 197  1.201417647   2  -6.04495062
## 198 -2.502542723   2 -11.00214270
## 199 -1.222331833   1  -6.22402036
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## 201  4.397620698   3  -3.83589123
## 202  2.624825953   2  -3.99504015
## 203 -0.530290113   2 -10.99026705
## 204  1.086388118   0   1.38959887
## 205 -0.828679896   0  -1.36972765
## 206 -0.952493789   2  -9.63457305
## 207 -1.577205676   0  -4.37619547
## 208 -1.189234535   2  -7.48965304
## 209  3.301814935   2  -1.57916907
## 210 -0.108056250   1  -3.47467713
## 211  0.238490473   1  -5.33786026
## 212  0.487374859   2  -3.94040710
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## 215 -1.985014301   1  -7.23774404
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## 218 -1.446131940   2 -10.21598153
## 219 -2.472546238   2  -9.03890494
## 220 -2.569431445   1  -8.10769079
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## 223  2.219696278   3  -7.06759596
## 224  1.415176708   0  -1.15277385
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## 227 -1.409192927   0  -5.39143704
## 228 -1.434436323   1  -7.68460755
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## 230 -2.031185157   0  -4.73647040
## 231  3.910587931   2  -0.12668935
## 232 -0.180639188   3 -11.56821901
## 233  0.429077653   1  -3.56561526
## 234 -1.477055409   2  -8.36833282
## 235 -1.148777380   1  -7.19782053
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## 237 -0.365850777   4 -12.08858018
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## 239  0.648608688   0  -1.31263218
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## 241 -1.577243942   1  -6.86815477
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## 249 -1.329538871   1  -5.13257972
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## 251 -0.751205743   7 -21.95319328
## 252 -1.123752727   5 -15.69101925
## 253 -0.687834468   1  -5.69633864
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## 256 -0.177130224   1  -3.32383950
## 257  2.161598992   2  -3.73929311
## 258  1.261508231   1  -2.64188811
## 259 -0.227279791   0  -2.97443484
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## 261 -1.042234635   2  -9.72969902
## 262 -0.979740906   3 -12.82478629
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## 264  2.600397355   0   1.60596279
## 265  4.586157948   2   1.58567666
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## 267 -0.266301929   0  -2.20191638
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## 280  0.049349966   0  -2.08069977
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## 315 -0.704092913   2  -8.84734581
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## 365  1.635318893   1  -2.43733171
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## 586 -0.452443963   2 -10.15418532
## 587  0.762851659   3 -10.05632839
## 588 -1.567031578   3 -12.12515132
## 589  1.165982811   1  -3.84565922
## 590 -2.633020803   1  -9.08299805
## 591 -5.619549358   1 -11.41782839
## 592  0.929935982   2  -5.72250128
## 593  1.681079654   2  -5.89033311
## 594 -0.571690847   0  -3.26906320
## 595  1.008252509   0  -0.87334997
## 596 -2.311833050   3 -13.71686145
## 597 -0.254297214   1  -5.92006971
## 598 -3.883036752   2 -12.68805553
## 599  2.362361783   3  -5.75149878
## 600  3.719821723   1   0.83460242
## 601  2.148024521   1  -1.79291268
## 602 -0.054693933   2  -6.88875126
## 603 -0.066660685   1  -3.89940659
## 604 -3.032135246   3 -14.18550330
## 605  1.580770688   0   0.66888416
## 606 -0.421468361   0  -3.73562861
## 607 -1.313485851   2  -9.28853088
## 608 -2.824051585   0  -6.37397049
## 609 -0.599524993   2  -7.84367049
## 610 -1.698122270   1  -6.62005166
## 611 -0.794061041   2  -8.43530344
## 612 -2.435199976   1  -8.77549419
## 613  3.375178966   0   1.79553634
## 614 -0.032005054   1  -4.36628647
## 615  2.149890155   1  -0.23331693
```

```
## 616 -5.203399341   2 -15.27189467
## 617 -0.906395660   0  -3.65375260
## 618 -1.350964583   2  -8.76066986
## 619 -2.445852359   1  -8.57881375
## 620  3.093218308   4  -7.94792205
## 621 -2.830563840   3 -13.81038549
## 622  0.636780521   0  -1.88475886
## 623  1.692872571   1  -1.39853715
## 624  0.356380377   2  -5.85475964
## 625 -1.750510955   2  -8.62657583
## 626  1.882331613   1  -0.69673915
## 627  0.341176164   1  -4.62871516
## 628 -2.126995813   0  -5.44227603
## 629 -2.776098095   1  -8.22531933
## 630  4.173434869   0   4.47288979
## 631 -1.357006291   2  -9.43943496
## 632 -3.711143308   2 -12.81894544
## 633  1.066518726   1  -3.72514531
## 634  0.620460512   1  -2.24915339
## 635 -2.707668671   0  -8.29890249
## 636 -3.885912827   0  -7.60785599
## 637 -0.232605034   3 -10.08743417
## 638  2.278792583   2  -5.06929999
## 639  1.272248070   2  -5.03342681
## 640 -0.985874848   1  -7.94731247
## 641 -1.668376463   1  -5.98075819
## 642  0.542133514   4 -13.30437367
## 643  0.314706694   0  -3.76214551
## 644  1.259423493   0  -1.00037094
## 645 -0.791595901   4 -13.34615387
## 646  1.798708106   2  -5.33337582
## 647 -1.661623055   7 -23.08555927
## 648 -0.661089400   1  -5.19232019
## 649  1.481629047   2  -6.18462026
## 650  1.979943217   0   0.68607096
## 651 -3.877009403   2 -12.12988017
## 652  0.214380814   3  -7.86587720
## 653  1.217558023   3  -7.61596689
## 654 -2.901648627   0  -4.34754509
## 655  0.961251200   1  -2.80108829
## 656 -1.656348541   2  -9.52212908
## 657  2.040506024   1  -3.99982642
## 658  1.076964064   2  -6.91690886
## 659  1.538104590   0  -0.16561620
## 660  0.241438659   2  -6.14971542
## 661  1.727296868   1  -2.50429796
## 662  2.761029057   1  -0.19091070
## 663  3.932496031   2  -2.75831691
## 664 -0.056790109   0  -2.35363587
## 665 -4.498102181   1 -10.68033702
```

```
## 666  0.063052009   2  -7.47111291
## 667  0.411122417   1  -3.91159692
## 668 -0.310690694   1  -5.70571500
## 669  1.136577242   1  -2.63444664
## 670  2.021355929   0   0.74884341
## 671 -1.035964860   0  -3.49740015
## 672 -0.588190654   0  -4.77225400
## 673  0.795684418   2  -6.75971040
## 674 -1.100447483   0  -5.03027056
## 675  0.182534767   0  -3.35016938
## 676 -3.923415198   0  -7.98666438
## 677 -2.239799443   1  -7.89651759
## 678 -2.655510951   3 -13.23411511
## 679 -1.707247393   1  -5.89888370
## 680 -1.386609056   1  -6.09685238
## 681  0.764610285   1  -2.51747526
## 682  1.964226005   1  -2.37639388
## 683 -1.454767058   3 -12.14451405
## 684 -1.993677965   3 -11.50598027
## 685 -2.083377718   1  -7.51552162
## 686 -0.829177453   2 -10.04109315
## 687 -0.478058136   0  -3.51583691
## 688  0.967235068   4 -10.03827885
## 689 -0.642649684   0  -3.51044682
## 690 -4.156978538   5 -20.48632582
## 691 -0.182868551   0  -2.24179328
## 692  2.374373621   1  -1.77223454
## 693  2.383202537   0   0.98569716
## 694 -1.577926436   2  -9.01737646
## 695 -3.095553087   2 -11.32419549
## 696  4.916120984   6  -9.48276048
## 697 -0.324843885   0  -2.65460122
## 698 -0.194902500   3 -10.99607154
## 699  0.841148377   2  -5.85648917
## 700 -3.228078913   5 -18.81397593
## 701 -1.456438222   1  -4.35704656
## 702 -3.080884811   2 -11.97692175
## 703 -1.386189228   0  -4.26701849
## 704  0.237698866   2  -5.98472838
## 705 -2.729418916   0  -4.53990036
## 706  1.179965359   3  -8.80673035
## 707  0.578688058   3  -8.32526626
## 708 -1.808430051   3 -12.89695959
## 709  0.452649885   3 -10.12846843
## 710  1.496162325   1  -1.88201184
## 711  2.122190507   4  -9.32243552
## 712 -0.425696557   1  -4.80664430
## 713 -0.187273588   1  -5.44268652
## 714 -0.173428270   1  -3.04782162
## 715  2.882923512   0   2.70138093
```

```
## 716  2.250143785   3  -7.02420113
## 717  1.668803136   0   1.03907332
## 718 -0.574681600   1  -6.61906862
## 719  0.746482868   2  -6.29267411
## 720  0.806580662   2  -4.66742917
## 721 -2.083346589   4 -15.50552454
## 722 -3.456609031   0  -5.92876227
## 723  1.283660055   1  -3.02479792
## 724 -3.058621062   3 -16.38009589
## 725  0.003367377   3 -10.67549822
## 726  0.500495643   3  -8.87657041
## 727  1.127734781   2  -4.17461860
## 728  0.378852475   0  -2.06300866
## 729 -1.465707613   0  -3.77601165
## 730  1.972731720   0  -0.20299488
## 731  3.477267534   1  -0.74828169
## 732  1.762357618   2  -5.51001598
## 733 -3.887301803   3 -13.87316046
## 734  2.799152371   1   0.25912341
## 735 -0.112111892   3  -9.36329149
## 736  1.049828559   1  -2.01477746
## 737  1.244066472   2  -5.88273194
## 738 -0.193372146   2  -6.71114407
## 739 -0.150526395   2  -7.20110768
## 740  2.038314139   3  -7.99546113
## 741  1.423203845   1  -3.43857684
## 742  1.980524492   0   0.82278722
## 743  4.765853390   0   4.87782481
## 744  1.328831727   0  -0.46099400
## 745  0.414762315   3 -10.21582400
## 746 -4.421266221   2 -15.48066465
## 747  5.383428006   0   5.96203469
## 748 -0.965353644   0  -3.64676681
## 749  4.749469430   0   5.19841809
## 750  0.749287136   1  -5.09688982
## 751  3.076860398   2  -4.14573138
## 752 -0.219420642   0  -2.41852077
## 753  1.022941509   0  -1.98069542
## 754  0.427915959   1  -5.98387151
## 755 -0.372241399   1  -4.33754255
## 756 -0.240787649   1  -4.03623337
## 757  2.025668672   0  -0.19402750
## 758 -0.402916294   3 -10.63614008
## 759 -4.075364989   2 -14.03948389
## 760 -0.391778498   1  -4.83206298
## 761  1.079581211   2  -5.71826461
## 762  1.232911433   1  -3.74338445
## 763  1.233135635   2  -6.24782765
## 764 -3.384203042   1 -10.39996545
## 765  0.737484116   2  -5.73114798
```

```
## 766  1.935718420   1  -1.31898536
## 767  2.553157363   2  -3.24910480
## 768 -0.449922542   1  -6.07278126
## 769 -0.643785172   2  -9.88746887
## 770  2.975675664   2  -2.50317841
## 771 -3.335856092   0  -6.84725748
## 772 -0.873659954   1  -5.86251629
## 773  0.914924159   1  -2.28697819
## 774 -3.235547531   1 -10.02328238
## 775  0.559255725   1  -2.67302666
## 776  3.755728042   3  -4.95517224
## 777 -0.008121307   2  -7.92455765
## 778 -0.556908050   0  -2.98541741
## 779  0.949823428   0  -0.28022747
## 780 -0.558144342   1  -5.58861772
## 781  1.626800748   1  -3.22298617
## 782  1.808870929   2  -4.94920498
## 783  0.005383322   0  -4.53439445
## 784 -2.353384315   4 -16.20324972
## 785 -2.636441455   3 -13.03505182
## 786 -1.185994731   2  -9.02965240
## 787  1.594761001   3  -6.87900688
## 788 -3.916410349   0  -6.99699206
## 789 -3.772650317   2 -11.83606370
## 790 -1.307559650   3 -10.97453856
## 791  0.788789696   0  -2.63427519
## 792 -1.827132096   1  -7.15382525
## 793  1.773498074   2  -3.08703536
## 794  0.666739940   1  -4.14704931
## 795 -0.341279236   3  -9.82498849
## 796  1.637656273   1  -2.16388749
## 797  0.776730326   2  -5.55556532
## 798 -0.891870054   2  -8.72064819
## 799  0.462229869   1  -4.60117748
## 800  1.295026716   3  -7.91500175
```

```
test=data[-(1:800),]
test
```

```
##               x1i x2i           yi
## 801   0.712566690   0  -0.79459180
## 802  -1.316020416   1  -6.62776087
## 803   1.710404410   1  -2.17399316
## 804   2.305872451   0   1.63957337
## 805   0.552549113   2  -7.20261513
## 806   0.288209325   2  -5.65682929
## 807  -0.151250162   1  -2.61098996
## 808   4.322831693   3  -2.27083030
## 809   0.552631064   2  -6.88073436
## 810  -0.316588057   4 -14.66611055
```

```
## 811  -5.015835604   3 -16.76797189
## 812  -3.130563530   1  -8.10251572
## 813  -0.155346401   2  -6.63937826
## 814   0.412588090   3 -10.45212345
## 815   0.553744918   3 -10.13790563
## 816   1.643013561   0   0.43396362
## 817  -0.388304819   0  -1.91754540
## 818   2.429177589   5 -10.72621971
## 819  -1.843032086   1  -6.36385294
## 820  -2.416885441   2 -11.55161511
## 821  -2.457972355   1  -9.35376635
## 822   1.484594048   1  -4.53408811
## 823  -0.165839885   2  -6.92724314
## 824   1.579635844   3  -6.76039825
## 825  -0.535412843   0  -2.41372729
## 826  -1.183784204   1  -7.31642233
## 827  -0.736705150   2  -6.66367046
## 828  -3.705233648   0  -7.55747373
## 829  -2.339230521   2  -8.69588437
## 830  -2.884069293   1  -8.08629500
## 831   2.108644544   2  -3.05758699
## 832  -1.194660183   2  -8.54192203
## 833   1.578919706   2  -5.05416464
## 834   3.032981203   3  -6.56637438
## 835  -0.383549618   1  -5.69548366
## 836   0.567757813   1  -3.85224690
## 837  -3.502135039   3 -14.42091499
## 838  -1.637339555   1  -6.92549459
## 839   0.112429699   2  -7.22057041
## 840   0.598173807   3  -8.77907115
## 841  -1.518796235   1  -7.25068677
## 842   5.369717998   1   2.38166774
## 843  -0.916780278   3 -12.53919709
## 844   0.128487113   2  -6.17384657
## 845   1.299583734   2  -5.33365040
## 846  -0.052037270   2  -7.35647641
## 847  -1.287134774   2  -9.74290730
## 848   2.090611328   1  -1.92091800
## 849   3.231090646   1  -1.16325595
## 850  -0.059387942   0  -3.13244047
## 851   1.124534691   4 -12.74460320
## 852  -0.194824997   1  -3.89105622
## 853   2.032910435   1  -1.62796131
## 854  -2.312334789   0  -6.35404508
## 855   4.641720448   2   0.46219523
## 856  -1.207062496   1  -5.66529581
## 857  -2.917698829   1  -7.94239609
## 858  -0.701835654   3 -11.00834048
## 859   0.293416968   1  -4.61208825
## 860   3.247242415   2  -2.75191352
```

```
## 861   1.822419354   2  -4.24392225
## 862   0.284916864   0  -0.80120178
## 863  -2.778967048   0  -4.30108472
## 864  -1.732075489   1  -7.53397609
## 865  -0.326569864   2  -8.67276256
## 866   5.106052227   2   0.04933178
## 867  -3.720455148   2  -9.38653302
## 868   2.262109304   3  -7.06166282
## 869  -1.054468514   0  -2.33057470
## 870   3.331981800   1  -0.17832369
## 871  -2.278401280   2 -10.86830503
## 872   0.287246469   0  -2.30827207
## 873  -2.199101883   0  -5.28998726
## 874   1.807032853   0   2.17163046
## 875   2.967558977   2  -2.74357071
## 876   3.901442013   3  -3.85065206
## 877   1.595201315   1  -1.52798237
## 878   3.686532509   1   2.63288703
## 879   2.492847817   1  -0.33439421
## 880  -0.263749824   4 -14.19636111
## 881   0.954074477   1  -4.29780707
## 882  -1.943988423   2 -11.51103891
## 883  -0.370404339   3 -13.16711284
## 884   2.441927417   2  -2.49056769
## 885   1.082568287   1  -3.57812827
## 886   0.914714663   3  -8.12442879
## 887  -2.076262079   2  -8.92514906
## 888  -1.209026461   0  -4.20446974
## 889  -1.529212028   2  -9.47586449
## 890   0.790591740   3  -8.34291538
## 891  -1.981015261   0  -4.54754214
## 892   1.124082780   4 -11.60059936
## 893  -2.232832824   0  -4.39515034
## 894   3.657060916   0   3.66552081
## 895   0.921182699   0  -0.99879352
## 896  -1.402007225   1  -7.78519103
## 897   0.482091865   2  -5.89173779
## 898  -0.704906398   2  -6.75934930
## 899   0.742295920   1  -2.18026822
## 900   0.487065438   0  -0.44547382
## 901  -2.028228345   4 -15.72265475
## 902  -1.582627758   2  -9.94702578
## 903   0.599187369   2  -6.94882244
## 904   3.278103818   1  -0.42235252
## 905   2.169234017   2  -3.45388675
## 906  -1.249134947   2  -8.69239217
## 907   1.651845803   3  -9.34377646
## 908  -0.097136706   1  -6.59657902
## 909   0.602627303   1  -3.77915587
## 910   0.520722981   2  -6.32104908
```

```
## 911   5.150899527   2  -2.29621500
## 912  -2.370577621   1  -8.73525590
## 913   0.201839718   2  -6.88986324
## 914  -3.559954576   1  -8.12215049
## 915   1.179671846   3 -10.16132814
## 916   2.193216943   2  -5.38494056
## 917   2.891324482   2  -4.23245257
## 918  -3.850290504   2 -12.41501063
## 919   0.825538994   2  -5.71340619
## 920   3.186739903   2  -3.05925337
## 921  -0.828031726   0  -4.77157310
## 922  -0.424301064   2  -8.42457391
## 923  -0.073074444   2  -7.86279021
## 924   0.730037503   1  -3.78052878
## 925   1.330319751   2  -3.71466717
## 926   2.635641767   0   1.01312794
## 927  -0.190975181   2  -7.39110085
## 928   0.392556091   4 -12.55581981
## 929   4.975995754   2  -1.47383379
## 930   0.862197856   2  -5.35697510
## 931   0.377506219   1  -5.12211997
## 932  -2.684486251   0  -3.02307156
## 933   0.005712095   1  -4.49944013
## 934  -0.442652306   0  -2.55945965
## 935  -0.022091660   2  -7.29747353
## 936  -1.150835283   1  -4.36752386
## 937  -1.373631305   2  -8.45915672
## 938  -1.441547264   1  -6.86915681
## 939  -0.429009031   0  -3.76680245
## 940   2.736265296   1  -1.80795635
## 941   2.098173253   1  -2.44754675
## 942  -0.719950236   3  -9.59298190
## 943  -3.371832909   0  -6.90954765
## 944  -1.689166857   0  -5.12203222
## 945  -0.915521066   0  -2.48967002
## 946   0.207276008   4 -10.76463357
## 947  -1.325214551   1  -7.14983186
## 948   4.013361382   2  -2.02622661
## 949  -0.544535069   0  -2.41706491
## 950  -2.427888940   1  -8.00388895
## 951  -0.282523515   0  -1.98916645
## 952  -2.010755163   2  -8.30085870
## 953   0.312311415   3  -9.42315057
## 954   0.467267228   0  -1.62652815
## 955   0.711175223   1  -3.11894023
## 956  -3.243716518   3 -14.39054766
## 957   0.441422582   0  -0.75424332
## 958   0.620900161   2  -6.55471093
## 959  -2.842216895   1  -8.13558094
## 960   1.910731279   0  -0.88871692
```

```
## 961   1.568341758   3  -7.59041788
## 962   4.599238722   0   3.92271822
## 963   0.313405975   2  -7.95177421
## 964   0.093467055   1  -4.88221533
## 965   0.193171668   2  -6.55756472
## 966   0.139532462   0  -1.71231159
## 967  -3.696945551   3 -14.28224001
## 968  -3.342254117   0  -5.73825652
## 969  -0.155077934   0  -1.38864508
## 970  -1.162134761   1  -6.55147784
## 971   0.109473050   2  -8.37489393
## 972  -4.222416747   1 -10.79171842
## 973  -2.997396509   1  -7.62669968
## 974  -2.202966878   0  -5.20562999
## 975   1.972116442   1  -3.47632863
## 976  -2.196980014   1  -7.18438880
## 977  -1.599027907   5 -16.95681976
## 978   0.159747637   0  -2.17695666
## 979  -0.645492724   3 -10.53032831
## 980   0.292834357   0  -0.22064423
## 981   4.610123964   2   0.55336964
## 982  -2.249207343   1  -8.20518717
## 983  -0.610939280   0  -2.84465013
## 984  -1.033518899   0  -4.90101560
## 985   3.024790854   0   1.50695387
## 986  -1.538969846   3  -9.94631739
## 987  -0.164173808   5 -13.73183365
## 988   1.574267229   1  -2.62528061
## 989  -2.117181072   0  -5.65651865
## 990   3.310351631   2  -3.93213101
## 991   1.351524831   2  -3.18181338
## 992  -2.148413219   1  -7.49314878
## 993   0.909155618   3  -9.12113104
## 994  -0.426614286   1  -4.11919266
## 995   0.626457544   3 -10.02254527
## 996  -0.179950394   0  -1.52536635
## 997   2.141032074   3  -5.36168479
## 998  -2.702200771   0  -5.99325124
## 999  -1.045233394   1  -4.61204581
## 1000 -0.498381356   2  -7.25618247
```

1. Fit a multiple linear regression equation of y on x1 and x2. Calculate test MSE.

```r
model=lm(yi~x1i+x2i)
print(model)
```

```
##
## Call:
## lm(formula = yi ~ x1i + x2i)
##
```

```
## Coefficients:
## (Intercept)          x1i          x2i
##      -2.045        1.371       -2.571
```

```r
yicap=predict(model,newdata = test)
testmse=mean((test$yi-yicap)**2)
testmse
```

```
## [1] 0.9913944
```

2. Fit a KNN model with k = 1, 2, 5, 9, 15. Calculate test MSE for each choice of k.

```r
#install.packages("caret")
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.5.2

## Loading required package: lattice
```

```r
#for k=1
tr_k_1=knnreg(yi~x1i+x2i,data=train,k=1)
Y_KNN=predict(tr_k_1,test)
mse_knn_1=mean((test$yi-Y_KNN)^2)
mse_knn_1
```

```
## [1] 2.219793
```

```r
tr_k_1=knnreg(yi~x1i+x2i,data=train,k=2)
Y_KNN=predict(tr_k_1,test)
mse_knn_1=mean((test$yi-Y_KNN)^2)
mse_knn_1
```

```
## [1] 1.729587
```

```r
tr_k_1=knnreg(yi~x1i+x2i,data=train,k=5)
Y_KNN=predict(tr_k_1,test)
mse_knn_1=mean((test$yi-Y_KNN)^2)
mse_knn_1
```

```
## [1] 1.303978
```

```r
tr_k_1=knnreg(yi~x1i+x2i,data=train,k=9)
Y_KNN=predict(tr_k_1,test)
mse_knn_1=mean((test$yi-Y_KNN)^2)
mse_knn_1
```

```
## [1] 1.205371
```

```r
tr_k_1=knnreg(yi~x1i+x2i,data=train,k=15)
Y_KNN=predict(tr_k_1,test)
mse_knn_1=mean((test$yi-Y_KNN)^2)
mse_knn_1
```

```
## [1] 1.23273
```

Suppose the data in Step (iii) is generated as :

1

yi = 1/(−2 + 1.4x1i − 2.6x2i + 2.9x 2 1i) + 3.1 sin(x2i) − 1.5x1ix 2 2i + εi . Work out the problems in (1) and (2). Compare and comment on the results.

```
set.seed(123)
x1i=rnorm(1000,0,2)
x1i

##    [1] -1.120951293 -0.460354979  3.117416628  0.141016783  0.258575470
##    [6]  3.430129974  0.921832412 -2.530122469 -1.373705704 -0.891323940
##   [11]  2.448163595  0.719627654  0.801542901  0.221365432 -1.111682270
##   [16]  3.573826274  0.995700956 -3.933234313  1.402711803 -0.945582815
##   [21] -2.135647412 -0.435949829 -2.052008897 -1.457782459 -1.250078536
##   [26] -3.373386621  1.675574089  0.306746236 -2.276273874  2.507629842
##   [31]  0.852928443 -0.590142966  1.790251322  1.756266975  1.643162163
##   [36]  1.377280508  1.107835307 -0.123823421 -0.611925327 -0.760942002
##   [41] -1.389413958 -0.415834556 -2.530792703  4.337911931  2.415923997
##   [46] -2.246217166 -0.805769671 -0.933310707  1.559930237 -0.166738133
##   [51]  0.506637028 -0.057093511 -0.085740915  2.737204568 -0.451541971
##   [56]  3.032941209 -3.097505608  1.169227499  0.247708488  0.431883137
##   [61]  0.759278966 -1.004646906 -0.666414767 -2.037150766 -2.143582453
##   [66]  0.607057283  0.896419557  0.106008453  1.844534936  4.100169371
##   [71] -0.982062332 -4.618337751  2.011477049 -1.418401525 -1.376017233
##   [76]  2.051142739 -0.569546014 -2.441435425  0.362606959 -0.277782725
##   [81]  0.011528372  0.770560802 -0.741320064  1.288753097 -0.440973124
##   [86]  0.663563928  2.193678026  0.870362982 -0.651863171  2.297615237
##   [91]  1.987007712  1.096793919  0.477463470 -1.255812152  2.721304897
##   [96] -1.200519174  4.374665986  3.065221252 -0.471400718 -2.052841801
##  [101] -1.420813127  0.513767418 -0.493383757 -0.695085199 -1.903237135
##  [106] -0.090055450 -1.569808939 -3.335883873 -0.760453041  1.837993218
##  [111] -1.150693925  1.215928644 -3.235765417 -0.111123931  1.038814408
##  [116]  0.602306724  0.211352388 -1.281412017 -1.699408692 -2.048257581
##  [121]  0.235293194 -1.894949228 -0.981114887 -0.512184384  3.687724010
##  [126] -1.303899803  0.470773145  0.155921699 -1.923713268 -0.142616172
##  [131]  2.889101717  0.903008106  0.082465844 -0.844993665 -4.106494443
##  [136]  2.262674427 -2.921280142  1.479895022  3.818207138 -2.887786322
##  [141]  1.403568671 -0.524394979 -3.144288318 -3.029335308 -3.203072347
##  [146] -1.061813044 -2.923511170  1.375833546  4.200217881 -2.574060952
##  [151]  1.575477695  1.538084482  0.664405158 -2.016753217 -0.238905213
##  [156] -0.560790670  1.125979066 -0.744877512  1.953946773 -0.749161716
##  [161]  2.105422931 -2.098354013 -2.520310490  6.482079870 -0.833715176
##  [166]  0.596455183  1.273139348 -0.967561251  1.033724089  0.737929055
##  [171] -0.430761015  0.130586067 -0.068134507  4.256903798 -1.482672193
##  [176] -2.191992534  0.075576798  0.620961499  0.873046958 -0.916730665
##  [181] -2.126652268  2.526370352 -0.699300776 -1.731025725 -0.472559138
##  [186] -0.394351789  2.219840579  0.169474584  1.508107570 -0.998584034
```

```
##  [191]  0.428890619 -0.649371823  0.189167056 -1.790726716 -2.621603067
##  [196]  3.994426769  1.201417647 -2.502542723 -1.222331833 -2.370960169
##  [201]  4.397620698  2.624825953 -0.530290113  1.086388118 -0.828679896
##  [206] -0.952493789 -1.577205676 -1.189234535  3.301814935 -0.108056250
##  [211]  0.238490473  0.487374859  2.464951757 -1.032127662 -1.985014301
##  [216]  3.351393865 -0.882326434 -1.446131940 -2.472546238 -2.569431445
##  [221] -1.147946959  1.235971634  2.219696278  1.415176708 -0.727314594
##  [226]  0.119499875 -1.409192927 -1.434436323  1.769300998 -2.031185157
##  [231]  3.910587931 -0.180639188  0.429077653 -1.477055409 -1.148777380
##  [236] -2.634032265 -0.365850777  0.837964810  0.648608688 -1.563072974
##  [241] -1.577243942 -1.004397437  2.992121340 -2.274607241 -0.358103189
##  [246]  3.804723643 -0.201949771 -2.719681408 -1.329538871  0.970919958
##  [251] -0.751205743 -1.123752727 -0.687834468  0.180993294  3.197017542
##  [256] -0.177130224  2.161598992  1.261508231 -0.227279791 -3.065804006
##  [261] -1.042234635 -0.979740906  0.094308866  2.600397355  4.586157948
##  [266]  3.095162118 -0.266301929 -3.513054791 -0.777559728  0.178414446
##  [271]  1.690026008  1.925055937  1.368618859 -2.790548700  1.699286091
##  [276] -0.893114433  0.349605400  0.149102354  0.856333530  0.049349966
##  [281] -3.334950195  1.472991930  0.772053137 -0.531303251  0.236289022
##  [286]  0.268077291  0.442038937  3.281692332 -0.438100758  0.336130768
##  [291]  2.336767746  2.108362047  2.290526221 -1.154936002  4.004965461
##  [296]  0.133401742  3.733703689 -2.701805372  0.041967173  2.499829142
##  [301] -1.430484374 -1.505377936 -1.877077407 -2.105026559 -0.874319066
##  [306]  0.662358346 -4.028420996  0.423960867  2.473350093  4.075148036
##  [311]  2.602351984  1.513549528 -3.453460798 -1.203013416 -0.704092913
##  [316]  1.407047806 -0.211342668 -2.517297256  3.368871416  1.822782584
##  [321]  0.474860545  2.436217221 -2.677548574  1.321640596 -1.045824753
##  [326]  1.367491044 -0.121643909  1.265921426  2.671035230  0.014580181
##  [331]  2.035117274 -2.376868070 -1.443208881  3.038435423  0.754775946
##  [336] -4.104445641 -2.728074904 -0.401562031  1.731558809 -0.203766511
##  [341]  1.248374944  1.918010756  3.342109658  0.112033467 -0.103963812
##  [346] -3.506474718  0.198655188 -1.143700116 -1.948019166 -0.359812462
##  [351]  2.029886345 -3.985496977 -0.854558574  0.233274567 -1.786415140
##  [356]  0.667805885  0.822859841 -0.066072319 -4.931796388  5.142916292
##  [361] -0.410598515  1.302386563  0.547532982  2.049346470  1.635318893
##  [366] -0.419586342  0.756335544 -1.890817662  1.713846022 -0.922076678
##  [371]  4.833546708 -3.302097791 -0.927974486  1.650759726  1.020265094
##  [376] -1.178962077 -1.993561484  0.288951409 -0.028614826 -3.580562475
##  [381]  0.069102134  0.380460631  0.349452794 -2.110034085  0.952266557
##  [386]  2.757140274  0.912472806 -2.271176941 -0.871290939  0.692207239
##  [391] -1.294091263 -4.315292670  1.768501640 -1.658955223 -1.147120542
##  [396]  3.007801218 -1.548289859  1.691463080 -2.521365758 -0.709084806
##  [401] -0.147112038 -2.337302849 -1.269496530 -0.057683106  1.341391937
##  [406] -3.301093087 -0.699508478  1.512812877 -1.077618320  0.454583843
##  [411]  0.984457140  0.535670031  1.306515359 -0.245417322 -0.827353027
##  [416] -5.286297904 -0.185882037  0.860569393  1.070797682 -1.110556703
##  [421]  3.559005820  0.572848839  0.252631717  2.544533559 -1.436932443
##  [426] -0.900677248  4.794904960  0.022258374  3.267136843 -2.877013290
##  [431] -0.381033604  0.756847807  0.600077090 -2.011272519  0.038518549
##  [436] -2.154841306  1.425406650  2.169550180 -4.449975393  2.471386925
```

```
##  [441] -2.482088994  0.909538538  1.319805276 -0.399779656 -1.290227914
##  [446]  0.330642042  0.877637401  1.766605640 -4.104673968 -3.272758536
##  [451]  2.860804682  2.093257694  0.870577898  1.430356814  1.834349835
##  [456] -5.321845597  2.220554193 -0.969975193  0.461233661 -0.590315602
##  [461]  1.743929908 -0.696944898  1.037007532 -0.781369957 -2.185574418
##  [466]  2.420021021  1.481800023  3.448524478  0.130307865  2.250005492
##  [471]  3.950838108 -0.562964230 -2.645902226 -0.478703134 -0.428082480
##  [476]  0.303361009  3.424609955 -0.652287785  0.746009312 -0.455368130
##  [481]  0.040901417  0.628115327  2.656429392  0.242636755  1.425684640
##  [486]  1.557720060  1.829546542 -1.148789104  3.253762428 -0.761913479
##  [491] -0.211568335  2.808100535  2.588167812 -2.179983745 -1.746142001
##  [496] -2.716158119  0.363694385  0.329681736  0.728229375  1.104315428
##  [501] -1.203785692 -1.987397182  2.053570112  1.502122605 -3.018333075
##  [506] -0.190294902 -1.791895646 -4.141502141  0.300240262 -0.158423419
##  [511] -0.194738535  0.432305084  1.764930329  0.411195009 -1.232871686
##  [516] -1.469598504 -0.263605586  0.620033973 -2.079360705 -0.368617738
##  [521]  1.934534520 -0.216560182 -1.396841335 -0.551890337  2.229297091
##  [526]  1.100087922  2.473351600  0.278195714  0.820550193 -1.116913825
##  [531]  1.210741338 -1.012667084 -2.841131009  0.255985931  3.891702435
##  [536]  1.601828679  2.330506780  0.717711446 -1.217114357 -0.404481710
##  [541] -0.546496214 -0.937399560  1.408334568 -2.394727005  1.732732264
##  [546]  1.728304972 -2.397244718  1.278983996  4.860453305 -1.114430964
##  [551]  1.689808483 -1.564403691  2.221422836  0.499649440  3.303830783
##  [556] -2.917941457 -0.102595772 -1.053850352 -0.394529738 -1.259157485
##  [561] -1.667687161  1.157444749 -2.175161428  2.968061863 -2.372413171
##  [566]  0.202158303  1.065978574  1.173470678 -0.603493328  0.159003991
##  [571]  1.922528304 -2.912931834 -1.563479423  0.640804629 -0.889563957
##  [576]  2.740007988  1.346507727  0.144333505 -3.015514635  0.052200451
##  [581] -0.632831736 -0.204693027 -2.363118454  0.997316088 -2.077912881
##  [586] -0.452443963  0.762851659 -1.567031578  1.165982811 -2.633020803
##  [591] -5.619549358  0.929935982  1.681079654 -0.571690847  1.008252509
##  [596] -2.311833050 -0.254297214 -3.883036752  2.362361783  3.719821723
##  [601]  2.148024521 -0.054693933 -0.066660685 -3.032135246  1.580770688
##  [606] -0.421468361 -1.313485851 -2.824051585 -0.599524993 -1.698122270
##  [611] -0.794061041 -2.435199976  3.375178966 -0.032005054  2.149890155
##  [616] -5.203399341 -0.906395660 -1.350964583 -2.445852359  3.093218308
##  [621] -2.830563840  0.636780521  1.692872571  0.356380377 -1.750510955
##  [626]  1.882331613  0.341176164 -2.126995813 -2.776098095  4.173434869
##  [631] -1.357006291 -3.711143308  1.066518726  0.620460512 -2.707668671
##  [636] -3.885912827 -0.232605034  2.278792583  1.272248070 -0.985874848
##  [641] -1.668376463  0.542133514  0.314706694  1.259423493 -0.791595901
##  [646]  1.798708106 -1.661623055 -0.661089400  1.481629047  1.979943217
##  [651] -3.877009403  0.214380814  1.217558023 -2.901648627  0.961251200
##  [656] -1.656348541  2.040506024  1.076964064  1.538104590  0.241438659
##  [661]  1.727296868  2.761029057  3.932496031 -0.056790109 -4.498102181
##  [666]  0.063052009  0.411122417 -0.310690694  1.136577242  2.021355929
##  [671] -1.035964860 -0.588190654  0.795684418 -1.100447483  0.182534767
##  [676] -3.923415198 -2.239799443 -2.655510951 -1.707247393 -1.386609056
##  [681]  0.764610285  1.964226005 -1.454767058 -1.993677965 -2.083377718
##  [686] -0.829177453 -0.478058136  0.967235068 -0.642649684 -4.156978538
```

```
##  [691] -0.182868551  2.374373621  2.383202537 -1.577926436 -3.095553087
##  [696]  4.916120984 -0.324843885 -0.194902500  0.841148377 -3.228078913
##  [701] -1.456438222 -3.080884811 -1.386189228  0.237698866 -2.729418916
##  [706]  1.179965359  0.578688058 -1.808430051  0.452649885  1.496162325
##  [711]  2.122190507 -0.425696557 -0.187273588 -0.173428270  2.882923512
##  [716]  2.250143785  1.668803136 -0.574681600  0.746482868  0.806580662
##  [721] -2.083346589 -3.456609031  1.283660055 -3.058621062  0.003367377
##  [726]  0.500495643  1.127734781  0.378852475 -1.465707613  1.972731720
##  [731]  3.477267534  1.762357618 -3.887301803  2.799152371 -0.112111892
##  [736]  1.049828559  1.244066472 -0.193372146 -0.150526395  2.038314139
##  [741]  1.423203845  1.980524492  4.765853390  1.328831727  0.414762315
##  [746] -4.421266221  5.383428006 -0.965353644  4.749469430  0.749287136
##  [751]  3.076860398 -0.219420642  1.022941509  0.427915959 -0.372241399
##  [756] -0.240787649  2.025668672 -0.402916294 -4.075364989 -0.391778498
##  [761]  1.079581211  1.232911433  1.233135635 -3.384203042  0.737484116
##  [766]  1.935718420  2.553157363 -0.449922542 -0.643785172  2.975675664
##  [771] -3.335856092 -0.873659954  0.914924159 -3.235547531  0.559255725
##  [776]  3.755728042 -0.008121307 -0.556908050  0.949823428 -0.558144342
##  [781]  1.626800748  1.808870929  0.005383322 -2.353384315 -2.636441455
##  [786] -1.185994731  1.594761001 -3.916410349 -3.772650317 -1.307559650
##  [791]  0.788789696 -1.827132096  1.773498074  0.666739940 -0.341279236
##  [796]  1.637656273  0.776730326 -0.891870054  0.462229869  1.295026716
##  [801]  0.712566690 -1.316020416  1.710404410  2.305872451  0.552549113
##  [806]  0.288209325 -0.151250162  4.322831693  0.552631064 -0.316588057
##  [811] -5.015835604 -3.130563530 -0.155346401  0.412588090  0.553744918
##  [816]  1.643013561 -0.388304819  2.429177589 -1.843032086 -2.416885441
##  [821] -2.457972355  1.484594048 -0.165839885  1.579635844 -0.535412843
##  [826] -1.183784204 -0.736705150 -3.705233648 -2.339230521 -2.884069293
##  [831]  2.108644544 -1.194660183  1.578919706  3.032981203 -0.383549618
##  [836]  0.567757813 -3.502135039 -1.637339555  0.112429699  0.598173807
##  [841] -1.518796235  5.369717998 -0.916780278  0.128487113  1.299583734
##  [846] -0.052037270 -1.287134774  2.090611328  3.231090646 -0.059387942
##  [851]  1.124534691 -0.194824997  2.032910435 -2.312334789  4.641720448
##  [856] -1.207062496 -2.917698829 -0.701835654  0.293416968  3.247242415
##  [861]  1.822419354  0.284916864 -2.778967048 -1.732075489 -0.326569864
##  [866]  5.106052227 -3.720455148  2.262109304 -1.054468514  3.331981800
##  [871] -2.278401280  0.287246469 -2.199101883  1.807032853  2.967558977
##  [876]  3.901442013  1.595201315  3.686532509  2.492847817 -0.263749824
##  [881]  0.954074477 -1.943988423 -0.370404339  2.441927417  1.082568287
##  [886]  0.914714663 -2.076262079 -1.209026461 -1.529212028  0.790591740
##  [891] -1.981015261  1.124082780 -2.232832824  3.657060916  0.921182699
##  [896] -1.402007225  0.482091865 -0.704906398  0.742295920  0.487065438
##  [901] -2.028228345 -1.582627758  0.599187369  3.278103818  2.169234017
##  [906] -1.249134947  1.651845803 -0.097136706  0.602627303  0.520722981
##  [911]  5.150899527 -2.370577621  0.201839718 -3.559954576  1.179671846
##  [916]  2.193216943  2.891324482 -3.850290504  0.825538994  3.186739903
##  [921] -0.828031726 -0.424301064 -0.073074444  0.730037503  1.330319751
##  [926]  2.635641767 -0.190975181  0.392556091  4.975995754  0.862197856
##  [931]  0.377506219 -2.684486251  0.005712095 -0.442652306 -0.022091660
##  [936] -1.150835283 -1.373631305 -1.441547264 -0.429009031  2.736265296
```

```
##  [941]  2.098173253 -0.719950236 -3.371832909 -1.689166857 -0.915521066
##  [946]  0.207276008 -1.325214551  4.013361382 -0.544535069 -2.427888940
##  [951] -0.282523515 -2.010755163  0.312311415  0.467267228  0.711175223
##  [956] -3.243716518  0.441422582  0.620900161 -2.842216895  1.910731279
##  [961]  1.568341758  4.599238722  0.313405975  0.093467055  0.193171668
##  [966]  0.139532462 -3.696945551 -3.342254117 -0.155077934 -1.162134761
##  [971]  0.109473050 -4.222416747 -2.997396509 -2.202966878  1.972116442
##  [976] -2.196980014 -1.599027907  0.159747637 -0.645492724  0.292834357
##  [981]  4.610123964 -2.249207343 -0.610939280 -1.033518899  3.024790854
##  [986] -1.538969846 -0.164173808  1.574267229 -2.117181072  3.310351631
##  [991]  1.351524831 -2.148413219  0.909155618 -0.426614286  0.626457544
##  [996] -0.179950394  2.141032074 -2.702200771 -1.045233394 -0.498381356
```

```
x2i=rpois(1000,1.5)
x2i
```

```
##    [1] 0 0 0 1 1 2 1 0 0 0 3 1 2 2 5 0 2 1 1 0 6 1 6 0 0 1 2 2 1 2 2 2 2 0
0 2 2
##   [38] 4 4 2 1 0 1 2 3 3 3 0 0 1 2 0 4 0 3 0 2 2 2 2 0 3 2 3 0 0 1 1 3 3 0
3 2 1
##   [75] 2 2 1 1 4 0 0 2 2 0 1 2 1 5 2 1 0 1 0 0 1 2 4 1 1 0 2 1 1 4 2 2 0 0
3 3 2
##  [112] 1 0 1 1 0 2 0 2 3 3 2 1 0 3 0 3 1 1 3 0 1 0 0 2 2 1 3 4 1 2 0 1 0 0
0 1 1
##  [149] 2 2 1 4 2 0 1 1 1 3 1 1 4 0 2 2 2 6 0 5 2 1 0 1 0 1 3 0 1 3 0 2 4 2
0 1 2
##  [186] 0 3 2 1 2 2 0 2 3 3 2 2 2 1 0 3 2 2 0 0 2 0 2 2 1 1 2 0 0 1 0 0 2 2
1 1 2
##  [223] 3 0 5 0 0 1 3 0 2 3 1 2 1 1 4 1 0 4 1 2 0 4 3 3 1 2 1 1 7 5 1 2 1 1
2 1 0
##  [260] 5 2 3 1 0 2 0 0 3 0 1 1 1 0 0 3 4 4 1 1 0 1 3 1 1 1 0 1 0 2 1 1 3 0
1 3 1
##  [297] 1 3 2 3 1 3 0 2 2 0 2 2 3 1 1 0 1 2 2 3 0 2 0 2 2 1 1 2 1 1 1 0 1 1
0 0 2
##  [334] 2 2 0 0 4 3 3 2 0 3 3 0 0 2 1 1 1 2 3 1 2 1 0 0 1 0 1 1 2 1 3 1 1 1
2 2 2
##  [371] 3 1 0 1 1 1 3 1 0 4 1 0 4 1 5 0 2 2 0 2 2 0 1 1 3 5 0 1 2 3 2 2 1 3
2 1 1
##  [408] 2 2 1 3 2 0 0 3 0 1 1 2 1 2 3 1 1 2 1 0 3 1 2 2 1 0 2 0 1 0 2 0 2 2
1 0 0
##  [445] 0 3 1 0 1 2 2 1 0 2 3 5 2 1 2 1 0 2 4 0 1 1 0 0 1 0 0 0 1 2 1 1 2 0
3 1 1
##  [482] 1 2 1 3 0 0 2 1 2 2 1 2 3 2 3 3 1 2 0 3 2 2 1 2 1 2 2 3 1 0 1 0 1 2
2 1 2
##  [519] 1 1 2 2 2 0 1 4 1 0 1 4 3 1 2 1 3 0 1 2 3 3 3 0 2 2 2 3 1 4 1 0 2 0
1 1 3
##  [556] 2 1 0 1 1 1 3 1 1 1 3 0 1 0 3 1 1 1 1 0 0 0 0 1 2 2 0 4 0 1 2 3 3 1
1 1 2
##  [593] 2 0 0 3 1 2 3 1 1 2 1 3 0 0 2 0 2 1 2 1 0 1 1 2 0 2 1 4 3 0 1 2 2 1
1 0 1
```

```
##  [630] 0 2 2 1 1 0 0 3 2 2 1 1 4 0 0 4 2 7 1 2 0 2 3 3 0 1 2 1 2 0 2 1 1 2
0 1 2
##  [667] 1 1 1 0 0 0 2 0 0 0 1 3 1 1 1 1 3 3 1 2 0 4 0 5 0 1 0 2 2 6 0 3 2 5
1 2 0
##  [704] 2 0 3 3 3 3 1 4 1 1 1 0 3 0 1 2 2 4 0 1 3 3 3 2 0 0 0 1 2 3 1 3 1 2
2 2 3
##  [741] 1 0 0 0 3 2 0 0 0 1 2 0 0 1 1 1 0 3 2 1 2 1 2 1 2 1 2 1 2 2 0 1 1 1
1 3 2
##  [778] 0 0 1 1 2 0 4 3 2 3 0 2 3 0 1 2 1 3 1 2 2 1 3 0 1 1 0 2 2 1 3 2 4 3
1 2 3
##  [815] 3 0 0 5 1 2 1 1 2 3 0 1 2 0 2 1 2 2 2 3 1 1 3 1 2 3 1 1 3 2 2 2 2 1
1 0 4
##  [852] 1 1 0 2 1 1 3 1 2 2 0 0 1 2 2 2 3 0 1 2 0 0 0 2 3 1 1 1 4 1 2 3 2 1
3 2 0
##  [889] 2 3 0 4 0 0 0 1 2 2 1 0 4 2 2 1 2 2 3 1 1 2 2 1 2 1 3 2 2 2 2 2 0 2
2 1 2
##  [926] 0 2 4 2 2 1 0 1 0 2 1 2 1 0 1 1 3 0 0 0 4 1 2 0 1 0 2 3 0 1 3 0 2 1
0 3 0
##  [963] 2 1 2 0 3 0 0 1 2 1 1 0 1 1 5 0 3 0 2 1 0 0 0 3 5 1 0 2 2 1 3 1 3 0
3 0 1
## [1000] 2
```

```
ei=rnorm(1000,0,1)
ei
```

```
##    [1] -0.820986697 -0.307257233 -0.902098009  0.627068743  1.120355028
##    [6]  2.127213552  0.366114383 -0.874781377  1.024474863  0.904758894
##   [11] -0.238248696 -1.557854904  0.761309895  1.129144396 -0.295107831
##   [16]  0.536242818 -0.275890475  0.682315245 -0.117290715 -0.344675864
##   [21]  0.111620498 -0.283405315 -0.591017164 -0.315936931 -0.008152152
##   [26]  0.207495141  1.532423622 -1.357997831 -0.199619051  0.631523128
##   [31]  1.762020903  0.426014363 -0.013753416 -0.307556910  0.414308164
##   [36]  0.989057920 -0.183858311  0.163761407  0.216936344  0.729277634
##   [41]  1.111380407  0.279160817 -0.076170672  1.394663132  0.164534118
##   [46]  1.577851979 -0.061922658  0.613922964 -1.546088594 -0.112391961
##   [51] -0.021794540 -0.758345417 -1.035892884  0.948159303  0.914158734
##   [56] -1.298731995  0.424378795 -1.112545320 -1.051073226  0.525412448
##   [61] -0.686024000  0.993479982  0.038523599  0.536148976 -0.523626698
##   [66] -1.151221335  0.914752241  0.238071492 -0.239067759  0.069235327
##   [71]  1.325908343 -0.698166635 -0.749408444 -0.619615053 -1.584991268
##   [76]  0.819628138  0.192369647  0.207171974 -0.043347354 -0.510160441
##   [81] -0.823418614  0.851856403 -1.426184673  0.440298942 -0.792611651
##   [86]  0.282310215 -0.740690522 -0.523341683  1.769365917  0.668282619
##   [91] -2.144897024  0.126412416 -0.451812936 -1.136626188  0.209785890
##   [96]  0.129965516 -0.328506573  1.972703567 -2.248690067  0.838219387
##  [101] -0.289023270  0.656513411 -0.453997701 -0.593864562 -1.710379666
##  [106] -0.209448428  2.478745801  0.989702208  1.675572156  0.914965318
##  [111]  1.144262708  0.902876414  0.475392432 -0.582528774 -0.532934737
##  [116] -1.600839996 -0.005817714  0.899355676  1.031922557  0.095132704
##  [121] -0.547627617  3.290517443  0.736685531  1.420575305 -0.337680641
```

```
##  [126] -0.037957627  0.448607098  1.676522312 -0.311474545  0.853615667
##  [131] -2.094814634 -0.507254434 -1.292009077  1.113362717 -0.164453088
##  [136] -0.390374082  1.369099846  1.116272858 -0.898021203  0.427866488
##  [141] -1.228444569 -0.475615024  1.616577637  1.450127951  1.109018755
##  [146] -0.570903886 -1.881431470 -1.175698184  0.952556525 -0.290567886
##  [151] -2.162608146 -0.180187488  1.410239221  0.643468641 -0.821258544
##  [156] -1.545916652 -0.826547226  0.034527671  0.888073701 -1.939940155
##  [161]  1.023201755  0.005457727  0.569778970 -1.653255563 -0.666654380
##  [166] -0.448234189  1.043891348  1.028174047  0.435090459  1.604212182
##  [171] -0.515411200  1.012537194 -0.035940030 -0.667342096  0.923380038
##  [176]  1.381100331  0.878250416 -0.509403455 -0.469787634  1.377675847
##  [181]  0.352826406  0.829573979 -0.338701984  1.261034936 -0.808755145
##  [186]  0.625351521 -0.817174966 -2.462575017 -1.342957511  0.136295199
##  [191]  0.882922750 -1.751302083 -1.251424469  1.764545997 -0.433899350
##  [196]  0.505700132 -0.526935321 -0.298582885  0.087244207  0.010961843
##  [201] -0.192560211 -0.469796485 -3.047860890  1.868655500  1.790424208
##  [206] -1.101081742 -0.168107522  1.375275305  0.998290023  1.276601619
##  [211] -1.071746918  2.577268096 -1.133459960  0.753916342  0.141275980
##  [216] -0.403710318 -0.379415798 -0.991396813  1.622659796  0.089513228
##  [221]  0.259217949  0.209632833 -0.375170753 -1.134021245  0.253726310
##  [226] -2.093639450 -1.418566943 -1.076396695 -1.078678863  0.107188817
##  [231]  1.598487548 -1.515324144  0.433676021  0.899544753 -0.989532202
##  [236] -0.052799402  0.823610903 -0.255509103 -0.220684347  0.307726791
##  [241] -0.060013253 -0.555652890 -0.138615019  1.882839792  0.873668680
##  [246] -0.914597073 -1.244917622 -0.359982241  1.328774701  0.292679118
##  [251] -0.701505237  0.882234568 -0.133370389 -1.120678499  0.461192454
##  [256]  1.524142810  0.434468298  0.192000371 -0.656243128  0.568398531
##  [261] -1.070570535 -1.653149024 -0.043352768 -0.034593506  2.365055532
##  [266] -1.216347308  0.170906323  0.805053094  1.050592844 -0.010724485
##  [271] -0.743256141 -0.065784052  1.939755992  0.482739008 -2.044477073
##  [276]  1.423459129  0.540502661 -0.033571772 -0.017863621 -0.149789720
##  [281]  0.256559481 -0.503866933  0.277011252 -0.931356025  0.200146875
##  [286]  1.106837421  0.509206114  1.033749676 -1.090868762  0.054792784
##  [291]  0.617250303 -1.068004868  1.565814337 -1.034808011  0.164518709
##  [296]  0.151832330  0.121670302 -0.210424584  0.449936787 -1.031164492
##  [301] -1.289364188 -0.654568638 -0.057324104  1.256747820  1.587454140
##  [306]  0.319481463  0.381591623 -0.243644884  0.048053084 -1.404545861
##  [311]  0.289933729 -0.535553582  0.334678773 -0.345981339 -0.661615735
##  [316] -0.219111377 -0.366904911  1.094578208  0.209208082  0.432491426
##  [321] -1.240853586  1.496821710  0.159370441 -0.856281403  0.309046645
##  [326]  0.870434030 -1.383677138  1.690106970 -0.158030705  1.121170781
##  [331]  0.072261319 -0.332422845 -1.834920047 -1.100172219 -0.041340300
##  [336]  0.827852545 -1.881678654  1.375441112  1.398990464 -1.143316256
##  [341]  0.472300562 -1.033639213 -0.125199979  0.928662739  0.868339648
##  [346] -0.849174604 -0.386636454 -0.976163571  0.339543660 -1.559075164
##  [351] -2.629325442  1.469812282  2.273472913 -0.455033540  0.761102487
##  [356] -0.007502784  1.474313800  0.554143933  0.203663965 -1.799136452
##  [361]  1.082955681 -0.350853615 -1.403490085 -0.201796665 -0.126778160
##  [366]  1.059206873 -1.167396032 -0.557643627  1.488119928  1.358665769
##  [371]  1.163214544  1.661523945  0.204030980 -0.581883687  0.555204062
```

```
##  [376]  1.058723126  2.413633271 -1.964982333  0.273235703  0.654794583
##  [381] -0.054598655 -1.557822248  0.741500892 -0.779085741  0.505861499
##  [386]  0.907551706  1.283957010 -1.557863797  1.081741848 -0.756981357
##  [391] -1.289019474  1.314320666  1.146259973 -0.242583268  0.759540706
##  [396] -0.860325741 -0.151031579 -0.093723234 -0.280740055  0.734098736
##  [401]  1.537327535 -0.455771063 -0.032658452  1.636757349 -0.329041971
##  [406] -2.604038171  0.513983786 -0.886468011 -0.998538406  1.420816814
##  [411]  2.447998007 -1.039782541  1.031025182 -0.094147838  0.141807459
##  [416]  1.222236701  0.213674518 -0.851365354 -0.470408875  0.686135255
##  [421] -2.335947332  1.095244382 -1.567150099  0.021931063 -0.190358978
##  [426]  1.293069486  0.188849315  0.101939135  0.698135809 -0.827014556
##  [431] -0.195898856  1.177584414  0.683473621 -1.275496707  0.637956367
##  [436] -1.377589625 -0.598310804  1.210920384 -2.251045175 -1.779014191
##  [441]  1.301372673 -0.814792784  1.243707019 -0.168250201  0.427775678
##  [446]  0.813278890 -0.651211869 -0.304590923 -0.415097170  2.816084276
##  [451]  0.126147069  0.472800423 -0.340753541 -0.241790643  1.378754669
##  [456] -0.338883670  0.020136303  0.376962164 -0.431723745  1.959064156
##  [461] -1.428459612  2.011292980 -0.351591889  1.357119647 -1.999177411
##  [466]  0.956080621  0.876431265 -1.271216972 -0.768323878  0.193524845
##  [471]  1.143835429 -0.765999304 -0.224125999  1.571346935 -1.127347241
##  [476]  0.947793981  0.448768189 -1.105814529 -0.667867842  0.783277510
##  [481]  0.248959433  1.425098277 -0.601783962 -1.714487696  1.047826926
##  [486] -0.608621618  0.120340527  1.719041814 -0.250414052  1.549555327
##  [491] -1.097139650  0.925511236  0.246799212 -0.736771539 -1.280008939
##  [496]  0.076643664  0.255164762  0.277446817  0.536856022 -0.460485566
##  [501] -0.511603722  0.236937879 -0.541589172  1.219227647  0.174135875
##  [506] -0.615268316 -1.806892964 -0.643681106  2.046018852 -0.560762421
##  [511] -0.835999306  0.652947503  0.441293125  0.751629057 -0.277975085
##  [516]  1.122654216 -1.172608860 -0.048876768 -0.704140337  0.680758638
##  [521]  0.130006761  1.109708082  2.058500869  0.140655532 -0.534616652
##  [526] -0.823516727 -0.263033982 -0.069601836  1.991801915 -1.129109543
##  [531] -1.093217437 -0.407966690  0.587559463  0.821111860 -0.907934699
##  [536]  0.127038606 -0.042892978  1.195206470  1.089192237 -0.312280694
##  [541]  0.045993767  0.652722613 -1.653492637 -0.310270965  0.574872884
##  [546] -0.523232154 -0.059918201 -0.021007535 -0.723653214 -0.994479836
##  [551] -0.199867235 -0.347027823  0.834095066  1.529882212 -0.011922379
##  [556]  0.398671992 -0.070415314  0.601359844  0.218495462  0.236595502
##  [561]  1.112915128 -0.987421149  1.447864013  0.349112412  0.180822011
##  [566] -0.560241851 -0.163877592  0.373684804 -2.063714257 -0.601521949
##  [571]  0.585991612 -0.294481786 -0.800527545 -0.635694531  0.235749032
##  [576] -1.634832385  0.871229241 -2.168934674 -0.503339522 -0.787182479
##  [581] -1.248600210 -1.077907340  0.250077354 -0.119774031 -0.300852632
##  [586] -2.320763776 -1.324320711 -0.131307112 -0.878035153 -0.796768930
##  [591]  1.049540713  0.175588349 -1.043844622 -0.468696015 -0.284903482
##  [596] -0.680295181 -0.964053610 -0.051804082  0.741194720  0.226852004
##  [601] -0.200147013  0.387820245  0.793918367 -0.140513958  0.455805199
##  [606] -1.145572907 -0.249650688 -0.420298275  0.195664504  0.357319514
##  [611] -0.123617979 -0.766214223 -0.929714217  0.278520611  1.356836852
##  [616] -0.787135595 -0.384798672  0.330680560 -0.554620450  0.121572315
##  [621] -0.047596117 -0.776251591  0.831441251  0.846307837  1.024139507
```

```
##  [626]  1.267996586 -0.506361788 -0.464481897  0.261218000  0.630080977
##  [631] -0.339626156 -0.423344808 -0.618271528  1.482201891 -2.508166352
##  [636] -0.167578034  0.038212877 -1.059609603  0.385425895 -1.967087684
##  [641]  0.954968861 -1.663360585 -2.202734880 -0.763563826  0.162080394
##  [646] -0.651567165 -0.559286995  0.333204975 -1.058900921 -0.085849546
##  [651]  0.497932993  1.633989657  0.479451881  1.714762992  0.453160034
##  [656] -0.003241127 -2.256534856 -1.224658552 -0.318962624  0.712270456
##  [661] -0.322513573  0.543648621 -1.063811352 -0.274129717  0.217006032
##  [666] -0.359385718  0.112831695 -0.670748026  0.374345223 -0.081054893
##  [671] -0.047049347 -1.948787086 -0.673668581 -1.489644085 -1.605718058
##  [676] -0.493883101 -0.160798368  0.283600226  1.091262650  0.444400297
##  [681]  1.012070341 -0.526310288 -0.307840173  1.085168884  0.001207184
##  [686] -1.680244716 -0.846555519  1.007592060 -0.610737258  0.333444133
##  [691]  0.014222696 -0.496357607 -0.350786392  0.391720548  0.209578829
##  [696]  1.234670140 -0.199819784 -0.923208042  0.165903102  0.705334553
##  [701]  2.281966956 -0.463683013 -0.326353572  0.882493210  1.281286127
##  [706] -0.658681857  0.664570455 -0.565157514 -0.962178270  0.623360902
##  [711]  0.106497772  0.389330878 -0.580503497  1.794977956  0.665288009
##  [716] -0.374402426  0.702748931 -1.214514384 -0.137750127  1.403357899
##  [721] -0.188839312  0.910490372 -0.221922000 -2.298026400 -0.880212546
##  [726]  0.222735689  1.446552711 -0.593402125  0.275979011 -0.964819289
##  [731] -1.016456241 -0.777316643  1.369062067  0.940310090  0.593665157
##  [736]  1.115462553 -0.424424997  0.759576937  0.209629278 -1.049100925
##  [741] -0.831062222  0.050052932  0.205630061 -0.321358419 -0.996491244
##  [746] -2.090891940  0.425235479 -0.295271709  0.549160883 -1.545891813
##  [751] -1.253335935 -0.111331875 -1.412813536 -1.982953851  0.783595408
##  [756]  0.900869340 -1.029963642 -0.272057269 -1.133972905  0.316426917
##  [761] -0.029678301 -0.869460451 -0.774217544 -1.062081193  0.436374262
##  [766]  0.571008848  0.376474894 -0.842889697 -1.786169628  0.530875663
##  [771] -0.177058947 -0.039392350  1.032127985 -0.893515834  1.144015326
##  [776] -0.413191495 -0.713187822 -0.205746144  0.390019729 -0.207215645
##  [781] -0.900507221 -0.281624279 -2.541931101 -0.508511682  0.455966218
##  [786] -0.169259775  0.688327722  0.485982434  0.645646746  0.656044945
##  [791] -1.738580761  0.004159680  1.630067335 -0.480485227  0.452802442
##  [796]  0.143393731  0.557012225 -0.272030117 -0.648299297  0.071960844
##  [801]  0.207814831 -0.185332290  0.031440665  0.411351934 -0.776183890
##  [806]  1.139677657  2.200760271  1.477205329 -0.454417851 -1.822887269
##  [811]  0.054197957  0.880273221  0.778106698 -1.229746775 -1.113148511
##  [816]  0.133744635  0.626081345  0.872931664  0.816391979 -0.967975492
##  [821] -1.312605056 -2.012519778  0.504932698  0.828111566  0.335850689
##  [826] -1.059124446  1.567716750 -0.370146624  1.779038361  0.551402005
##  [831]  1.190310645  0.330602226 -0.064652228 -1.012548068 -0.558514190
##  [836] -0.047107836  0.282074068 -0.033219212 -0.177971989  0.183485517
##  [841] -0.524372043 -0.535937456 -1.455704702  0.846271470  0.046932368
##  [846] -0.083624233 -0.740918615 -0.247773864 -1.086782856 -1.049297352
##  [851] -1.918951772  0.981698773  0.125964082 -1.116776376  1.163786604
##  [856]  0.624591683  0.742382275 -0.225770567 -0.422872007 -0.098052902
##  [861]  0.404690656  0.799914607  1.589469150 -0.509070404 -1.015564748
##  [866]  0.100858666  3.022104185 -0.428615845  1.145681218 -0.243098210
##  [871] -0.478543238 -0.710417122 -0.211244628  1.641784471  0.301846721
```

```
##  [876]  0.487329123  0.838735792  2.071741513  0.775618842 -1.427111353
##  [881] -1.033511335 -1.589455114 -2.848546770  1.290733928 -0.493723874
##  [886]  0.394970678  1.181617853 -0.511832695 -0.134967649  0.350256180
##  [891]  0.225879222 -0.774315250  0.730815610  0.545635528 -0.288449296
##  [896] -1.222380911  0.633333596  1.427519656  1.380517493  0.872634567
##  [901] -0.483135069 -0.531346919 -0.587684757 -0.411697869  0.709185621
##  [906]  0.256396754 -1.856360586 -1.860587630 -0.022834094  0.149938747
##  [911] -2.307474342 -0.816447226  0.027561152  1.461785915 -2.012868728
##  [916] -1.255444278 -1.080306847  0.175396079  0.330839221 -0.320689231
##  [921] -1.612328688 -0.630552417 -0.560485987 -0.202581284  1.622885181
##  [926] -0.676770530  0.076264405 -0.705398342 -1.240227849  0.635947898
##  [931] -1.050628680  2.735209190  0.092562938  0.060253576 -0.066545211
##  [936]  1.843645540  0.663927110 -0.250990644 -1.166189807 -1.038727761
##  [941] -0.784989305  1.214948431 -0.188981576 -0.757198623  0.792059478
##  [946]  1.345180019 -0.694531484 -0.444932544  0.345284187 -0.004844437
##  [951]  0.406366471  1.714198526 -0.060386554 -0.280702268  0.485414461
##  [956] -0.049344530  0.627765062 -0.223971151  0.443522714 -1.563740708
##  [961]  0.013903658 -0.516215987 -1.190542576 -0.413069208  0.371994945
##  [966]  0.092342964  0.693483763  0.940899243  0.828464030 -0.324489176
##  [971] -1.328156198 -0.280334970  1.169655437 -0.121476365 -1.637291651
##  [976]  0.491383223  0.281819311 -0.400603355  0.173361503  1.369387670
##  [981]  1.299196094 -0.456296894  0.010664862 -1.454089145 -0.727753326
##  [986]  2.008240397  1.498009686 -0.229254725 -0.692465145 -1.366623297
##  [991]  2.126051854  0.114629725 -0.593948909  1.078067338 -1.099585833
##  [996]  0.726564198  1.440870302 -0.210170160  1.451280944  0.641551431
```

```
yi1=1/(-2 + 1.4*x1i - 2.6*x2i + 2.9*x1i^2) +
        3.1*sin(x2i) -
        1.5*x1i*(x2i^2) + ei
yi1
```

```
##    [1]  1.258197e+01 -7.998900e-01 -8.693620e-01  2.793949e+00
3.093778e+00
##    [6] -1.560322e+01  4.086159e-01 -7.979896e-01  1.669925e+00 -1.546481e-
01
##   [11] -3.276015e+01 -5.070414e-01 -1.466392e+00  2.471581e+00
3.834323e+01
##   [16]  5.612161e-01 -3.772466e+00  9.219498e+00  7.129518e-01 -
1.712949e+00
##   [21]  1.144346e+02  2.764449e+00  1.092302e+02  1.553209e-01
1.271096e+00
##   [26]  7.918368e+00 -5.398034e+00 -5.335542e-01  5.961486e+00 -
1.152669e+01
##   [31] -7.933889e-01  6.643167e+00 -7.719088e+00 -2.012163e-01  5.373037e-
01
##   [36] -9.294120e-02 -4.490548e+00  7.096202e-01  1.247489e+01
7.961917e+00
##   [41]  4.747887e+00 -2.014455e-01  6.424445e+00 -2.179528e+01 -
3.191781e+01
##   [46]  3.293195e+01  1.114288e+01 -6.672492e-01 -1.407981e+00
```

```
2.535873e+00
##   [51] -4.168187e-01 -1.241326e+00 -1.404207e+00  9.906046e-01
7.345830e+00
##   [56] -1.264157e+00  2.188963e+01 -5.934670e+00  1.316911e-01  5.877678e-
01
##   [61]  6.747910e-01  1.487290e+01  6.709744e+00  2.685457e+01 -4.034969e-
01
##   [66] -1.343382e+01  1.193136e+00  2.461323e+00 -2.432532e+01 -
5.482320e+01
##   [71] -4.042175e-01  6.210880e+01 -9.863387e+00  2.785627e+00
9.214869e+00
##   [76] -8.541381e+00  3.430865e+00  6.585786e+00 -1.117888e+01 -9.720281e-
01
##   [81] -1.327584e+00 -1.179995e+00  5.690049e+00  6.567108e-01
2.262513e+00
##   [86] -1.080488e+00 -1.342175e+00 -3.622094e+01  8.354025e+00 -9.777136e-
02
##   [91] -2.063142e+00  3.447791e+00 -1.943384e+00  8.984240e-02 -
1.215269e+00
##   [96]  9.939187e+00 -1.076463e+02  2.055334e-02  8.503111e-01  9.743278e-
01
##  [101]  1.075482e+01  2.173421e+00  2.676526e+00  1.365856e+01
1.408995e+01
##  [106]  3.012768e+00  2.817874e+00  1.028763e+00  1.227032e+01 -
2.307137e+01
##  [111]  1.066609e+01  2.407020e+00  5.173503e-01  1.980842e+00 -
6.133606e+01
##  [116] -1.114941e+01  1.397279e+00  1.932550e+00  1.321663e+01
2.618815e+01
##  [121] -3.394025e+00  1.926320e+01  4.502656e+00  9.094044e-01 -
4.965575e+01
##  [126]  8.670288e-01 -5.587030e+00  3.819246e+00  5.473459e+00
3.115809e+00
##  [131] -2.056721e+00 -2.830089e-01 -1.828252e+00  2.143648e-01
2.732115e+01
##  [136] -1.105513e+01  8.421853e+00 -1.915111e+01 -9.485269e+01
7.432452e+00
##  [141] -4.739031e+00 -9.919621e-01  8.992412e+00  1.499215e+00
1.151995e+00
##  [146] -5.180428e+00  5.174534e+00 -2.757280e-01 -2.140987e+01
1.809151e+01
##  [151] -1.709098e+00 -3.973562e+01  4.221638e-02  7.869057e-01
1.935969e+00
##  [156]  1.680271e+00  1.624263e+00  1.041955e+01  6.743212e-01
1.543682e+00
##  [161] -5.155915e+01  1.331510e-01  1.864046e+01 -3.771883e+01
6.997015e+00
##  [166] -3.358656e+01  1.266958e+00  3.426574e+01 -3.325238e+00
2.602794e+00
##  [171] -9.996831e-01  3.196266e+00 -5.162646e-01 -4.425589e+00
```

```
2.119513e+01
##  [176]  1.493901e+00  3.150113e+00 -8.582913e+00  2.282061e-01
9.531491e+00
##  [181]  4.860423e+01 -1.144247e+01 -9.793750e-01  7.066277e+00
4.706802e+00
##  [186]  1.494112e-01 -3.021594e+01 -8.059611e-01 -7.530767e-01
8.771374e+00
##  [191]  9.635507e-01 -2.344338e+00  2.860122e-01  2.604434e+01
3.554999e+01
##  [196] -2.061965e+01 -5.667290e+00  1.766957e+01  4.023840e+00  1.020128e-
01
##  [201] -5.910390e+01 -1.333916e+01  2.812388e+00  2.208371e+00  9.347666e-
01
##  [206]  7.263283e+00  1.645729e-01  1.111958e+01 -1.595934e+01
3.835266e+00
##  [211]  9.352444e-01  2.300280e+00 -1.081025e+00 -2.057877e+00
5.974405e+00
##  [216] -3.753530e-01 -1.402322e+00  1.018774e+01  1.941825e+01
6.643557e+00
##  [221]  4.170509e+00 -5.349353e+00 -2.977195e+01 -9.612843e-01
2.448632e+01
##  [226] -2.651897e+00 -8.586633e-01  2.124118e+00 -2.395705e+01  2.476203e-
01
##  [231] -1.902276e+01  1.260358e+00  2.110051e+00  1.224068e+01
2.922236e+00
##  [236]  6.591319e+00  7.178095e+00  3.769445e-01  7.588001e+00
3.534211e+01
##  [241]  7.376351e+00  8.113516e+00 -1.030936e-01  5.240414e+01
6.044823e+00
##  [246] -5.181423e+01  1.456680e+00  1.887269e+01  5.182636e+00 -5.278234e-
01
##  [251]  5.649779e+01  3.997284e+01  3.268331e+00  4.662327e-01 -
1.691894e+00
##  [256]  4.188181e+00 -9.609654e+00  1.469724e+00 -1.117415e+00
1.126889e+02
##  [261]  7.820138e+00  1.189161e+01  2.198629e+00  1.246410e-02 -
2.231646e+01
##  [266] -1.183142e+00 -2.905262e-01  4.871622e+01  3.016673e-01
2.095357e+00
##  [271] -5.044180e-01 -2.317115e-01  2.126738e+00  5.427054e-01 -
2.349801e+01
##  [276]  2.042391e+01 -1.028265e+01  2.120216e+00  5.216008e-01 -6.695815e-
01
##  [281]  7.911052e+00 -2.064351e+01  1.168999e+00  2.253174e+00
2.210803e+00
##  [286]  4.007634e-01  2.161838e+00  1.063313e+00  4.218755e+00
1.896121e+00
##  [291] -2.104084e-01 -2.892793e+01  1.626710e+00  2.880383e+00 -
5.344141e+01
##  [296]  2.331018e+00 -2.845967e+00  3.683323e+01  2.876824e+00 -
```

```
3.425680e+01
##  [301]  1.968924e+00  1.991809e+01  1.215665e-01  1.707565e+01
9.491087e+00
##  [306]  5.329864e+00  2.740016e+01 -1.329212e-01 -3.281701e+01 -
4.888409e+00
##  [311] -9.515090e-01 -3.876767e-01  8.163189e+00  9.477575e+00
6.233573e+00
##  [316] -1.925554e+01 -8.285110e-01  1.914786e+01  2.372748e-01 -
7.484871e+00
##  [321] -1.441226e+00  5.134676e-01  6.864624e+00 -9.486262e+00
4.140596e+00
##  [326]  1.793044e+00  1.195816e+00  1.916366e+00 -1.505593e+00
3.489471e+00
##  [331]  1.500212e-01 -2.419737e-01  9.328712e+00 -1.646999e+01 -
1.973830e+00
##  [336]  8.521784e-01 -1.818242e+00  8.586807e+00 -2.078157e+01
1.944651e+00
##  [341] -5.271169e+00 -9.455617e-01 -4.476954e+01 -2.504105e-01  3.953486e-
01
##  [346] -8.143890e-01  1.093356e+00  2.932636e+00  6.142047e+00
1.377711e+00
##  [351] -1.185809e+01  5.574408e+01  5.892028e+00  8.152342e-01
6.513596e+00
##  [356]  4.374121e+00  2.370701e+00  3.048130e+00  2.198895e-01 -
6.892341e+00
##  [361]  4.094008e+00 -7.531438e+00  4.639495e-02 -2.723997e+01  2.124642e-
01
##  [366]  4.083328e+00 -2.246313e-01  1.552580e+01 -5.707133e+00
9.543980e+00
##  [371] -6.363674e+01  9.267877e+00 -1.043057e+00 -2.713240e-01 -
4.906840e+00
##  [376]  4.985212e+00  2.882571e+01 -4.280016e-02 -2.175170e-01
8.429280e+01
##  [381]  2.227562e+00 -2.512404e+00 -1.007798e+01  5.181181e+00 -
3.826740e+01
##  [386]  9.493835e-01 -1.657122e+00  1.510640e+01  9.969215e-02 -
2.297956e+00
##  [391]  9.053687e+00  1.336078e+00  1.246037e+00  5.799012e+00
1.655139e+01
##  [396] -1.165608e+02  2.081279e-01  1.425203e-01  1.779604e+01
1.063709e+01
##  [401]  5.102641e+00  1.657307e+01  3.893152e+00  2.751646e+00 -
1.517511e+01
##  [406]  5.000843e+00  3.931440e+00 -6.501389e+00  8.098763e+00
3.050263e+00
##  [411] -1.058292e+01 -1.612982e+00  1.240258e+00 -5.552072e-01
1.163710e+01
##  [416]  1.236195e+00  2.890975e+00 -3.352486e-01 -4.497298e+00
4.572648e+00
##  [421] -2.084219e+01 -6.325023e+00  4.162315e-01 -1.129936e+00
```

```
1.093987e+01
##  [426]  4.967616e+00  2.028574e-01  1.365417e-01 -1.561677e+00
1.933216e+01
##  [431]  4.772371e+00  2.118742e+00 -7.965266e+00  1.419393e+01  1.229627e-
01
##  [436]  4.634204e+00 -4.284664e-01 -8.882157e+00 -2.230719e+00 -
1.371694e+01
##  [441]  1.915179e+01 -6.486019e-01  1.447822e+00 -6.453033e-01
1.406943e+00
##  [446] -3.323781e+00 -2.381591e-01 -1.995913e-01  8.376439e+00
2.532333e+01
##  [451] -1.417117e+01  3.207358e-02  3.650942e-01 -4.645803e+00 -
2.255163e+01
##  [456]  1.962744e+02 -1.038641e+01  4.130839e+00 -5.487294e-01
5.226642e+00
##  [461] -1.320483e+00  8.864011e+00 -2.771358e+01  6.014644e-01
4.049224e+00
##  [466] -1.986893e-03  1.031659e+00 -1.244419e+00  1.415854e+00  2.566907e-
01
##  [471]  1.164328e+00 -1.301029e+00  6.436634e+00  7.123607e+00
1.909106e+00
##  [476]  2.845454e+00 -1.724843e+01 -1.701296e+00 -1.044155e+01
3.859195e+00
##  [481]  2.575800e+00  2.703363e+00 -1.366266e+01  2.855932e-01 -
1.828512e+01
##  [486] -4.700721e-01  2.177270e-01  1.122984e+01 -2.489879e+00
8.787956e+00
##  [491]  2.855341e+00 -6.330326e-01 -1.240029e+01  3.020601e+01
1.076950e+01
##  [496]  3.731058e+01 -4.329506e+00  2.129926e+00 -1.229097e+00 -1.360870e-
01
##  [501]  1.603967e+01  1.565953e+01 -9.917681e+00  1.821734e+00
2.116965e+01
##  [506]  2.068712e+00  9.244874e+00  2.705137e+01 -1.679423e+00
2.074862e+00
##  [511] -1.298393e+00  2.323430e+00  5.465082e-01  2.460430e+00
9.716745e+00
##  [516]  1.242509e+01  1.621607e+00 -1.141937e+00  5.222357e+00
3.630471e+00
##  [521] -8.501180e+00  5.092154e+00  1.297243e+01 -3.886254e-01 -
1.192683e+00
##  [526] -2.970776e+01 -1.304273e+00 -7.910574e-01  2.702270e+00
2.323408e+01
##  [531] -1.726023e+01  3.391057e+00  2.053493e+01  2.798876e+00 -
5.298317e+01
##  [536]  2.571869e-01 -8.607131e-01 -5.049434e-01  1.781897e+01
5.484601e+00
##  [541]  7.758061e+00 -6.560454e-01 -5.374628e+00  1.704144e+01 -
6.748419e+00
##  [546] -2.263787e+01  6.259326e+00 -3.323320e+01 -5.391632e+00
```

```
2.311926e+01
##  [551] -7.229750e+00 -3.051361e-03  1.885199e-01  3.074157e+00 -
4.413840e+01
##  [556]  2.079973e+01  2.479864e+00 -3.325616e+00  3.206127e+00
4.167299e+00
##  [561]  7.107444e+00 -1.640831e+01  7.483758e+00 -1.454584e+00
6.467038e+00
##  [566] -2.958307e+00  1.948449e-01  2.187051e+00 -2.622780e+00 -
2.415822e+00
##  [571]  4.242631e-01  6.746255e+00  7.485633e+00  6.135765e-01 -8.162737e-
01
##  [576] -1.592474e+00  1.065666e+00 -2.744468e+00  6.685476e+00
1.577968e+00
##  [581]  5.222799e+00 -1.539788e+00  5.667565e+01  3.186885e-01
5.624085e+00
##  [586]  3.074596e+00 -1.132730e+01  2.125586e+01  1.007221e+00
5.845933e+00
##  [591]  1.210006e+01 -2.880171e+00 -8.012903e+00 -1.008490e+00  1.388944e-
01
##  [596]  3.137298e+01  1.816242e+00  2.609740e+01 -3.061003e+01 -
2.719772e+00
##  [601] -7.287907e-01  3.397214e+00  3.288814e+00  4.131004e+01  5.898588e-
01
##  [606] -1.627521e+00  1.020230e+01 -3.620728e-01  6.468718e+00
6.235020e+00
##  [611]  7.305324e+00  5.603980e+00 -9.017512e-01  2.719656e+00  8.252090e-
01
##  [616]  3.326770e+01 -1.512894e+00  1.099203e+01  5.829966e+00 -
7.641094e+01
##  [621]  3.870806e+01  1.405788e+01  1.065142e+00  1.368938e+00
1.303760e+01
##  [626]  1.173389e+00  1.326219e+00 -3.416639e-01  7.106060e+00  6.484790e-
01
##  [631]  1.035524e+01  2.469864e+01  5.605184e+00  2.777653e+00 -
2.443527e+00
##  [636] -1.400682e-01  3.515539e+00 -1.182304e+01 -5.808801e+00
1.803986e+00
##  [641]  6.946093e+00 -1.711334e+01 -2.988779e+00 -5.343648e-01
1.672876e+01
##  [646] -8.412260e+00  1.235378e+02  3.698553e+00 -6.323682e+00 -3.480130e-
03
##  [651]  2.661111e+01 -9.294418e-01 -1.578352e+01  1.769246e+00 -1.203855e-
01
##  [656]  1.211378e+01 -2.611941e+00 -5.297049e+00 -1.763920e-01
1.933049e+00
##  [661] -1.503518e-01 -9.425468e-01 -2.181679e+01 -7.571857e-01
9.593649e+00
##  [666]  1.940283e+00  1.821764e+00  2.193545e+00  2.634062e+00 -2.184006e-
03
##  [671] -3.005598e+00 -2.498190e+00 -2.864247e+00 -3.624692e+01 -
```

```
2.212578e+00
##  [676] -4.669634e-01  5.954245e+00  3.671472e+01  6.944471e+00
4.097106e+00
##  [681]  1.928495e+00 -7.570074e-01  1.959353e+01  2.749777e+01
5.932049e+00
##  [686]  5.956582e+00 -1.344932e+00 -2.467215e+01 -1.198277e+00
1.532816e+02
##  [691] -4.489467e-01 -1.383016e+00 -2.946302e-01  1.222117e+01
2.166324e+01
##  [696] -2.650852e+02 -6.652038e-01  2.045073e+00 -2.314019e+00
1.188791e+02
##  [701]  5.023893e+00  2.090290e+01  2.864874e-01  2.125943e+00
1.344645e+00
##  [706] -1.639403e+01 -6.834955e+00  2.393495e+01 -6.752137e+00
1.238538e+00
##  [711] -5.289681e+01  3.422323e+00  2.098904e+00  4.453401e+00  7.035454e-
01
##  [716] -3.018939e+01  8.216190e-01  2.031187e+00 -2.018141e+00 -8.563025e-
01
##  [721]  4.709905e+01  9.464482e-01  9.672989e-01  3.950747e+01 -5.902904e-
01
##  [726] -6.215917e+00 -3.018366e+00 -1.542734e+00  7.350998e-01 -8.818157e-
01
##  [731] -3.595495e+00 -8.298691e+00  5.432010e+01 -6.044885e-01
2.443846e+00
##  [736]  1.730861e+01 -6.100958e+00  4.602804e+00  3.795463e+00 -
2.793288e+01
##  [741] -5.116631e-02  1.323716e-01  2.198062e-01 -1.206022e-01 -
6.272983e+00
##  [746]  2.727862e+01  4.363984e-01 -1.836192e+00  5.634332e-01 -5.813245e-
01
##  [751] -1.685496e+01 -5.726786e-01 -1.007414e+00 -3.044611e-01
3.738622e+00
##  [756]  3.660922e+00 -9.514436e-01  5.503706e+00  2.616540e+01
3.300041e+00
##  [761] -4.121497e+00  5.415036e-01 -6.294234e+00  6.664668e+00 -
1.387561e+00
##  [766]  3.873955e-01 -1.205820e+01  2.225169e+00  4.750423e+00 -
1.446020e+01
##  [771] -1.379977e-01  3.602619e+00  1.146667e+00  6.615469e+00
2.570051e+00
##  [776] -5.065055e+01  2.015688e+00 -7.375914e-01  9.038864e-01
3.015246e+00
##  [781] -5.453132e-01 -8.108613e+00 -3.043844e+00  5.635385e+01
3.663540e+01
##  [786]  9.556383e+00 -2.561526e+01  5.130109e-01  2.613510e+01
1.859570e+01
##  [791] -6.380521e-01  5.749707e+00 -5.965046e+00  7.073359e-01
5.396941e+00
##  [796]  4.782753e-01 -1.513749e+00  7.735195e+00  9.669107e-01 -
```

```
1.729359e+01
##  [801]  2.335148e+00  3.692979e+00  2.336686e-01  4.714204e-01 -
1.453128e+00
##  [806]  2.076703e+00  4.825466e+00 -5.642373e+01 -1.131866e+00
3.349473e+00
##  [811]  6.822326e+01  8.236123e+00  4.392906e+00 -6.476778e+00 -
8.274151e+00
##  [816]  2.567648e-01  1.513293e-01 -9.301252e+01  6.563979e+00
1.650949e+01
##  [821]  5.088404e+00 -1.372459e+00  4.182784e+00 -2.289810e+01 -1.854594e-
01
##  [826]  2.869198e+00  8.656562e+00 -3.394963e-01  1.881864e+01
7.550648e+00
##  [831] -8.527082e+00  1.010613e+01 -6.272950e+00 -4.147298e+01
2.413072e+00
##  [836]  1.361423e+00  4.804630e+01  6.164778e+00  1.823536e+00 -
7.580573e+00
##  [841] -2.283894e+01 -5.970398e+00  1.124264e+01  2.750745e+00 -
7.003338e+00
##  [846]  2.909775e+00  9.562476e+00 -6.842362e-01 -3.291746e+00 -
1.531710e+00
##  [851] -3.139357e+01  3.672531e+00 -2.170991e-01 -1.019393e+00 -
2.385153e+01
##  [856]  4.559387e+00  7.789980e+00  9.579578e+00  1.491726e+00 -
1.672688e+01
##  [861] -7.510317e+00  6.769007e-02  1.650056e+00  5.294496e+00
3.626584e+00
##  [866] -2.770340e+01  2.819972e+01 -3.040777e+01 -2.826754e+00 -
2.601514e+00
##  [871]  1.622507e+01 -1.446483e+00 -9.946032e-02  1.741790e+00 -
1.464023e+01
##  [876] -5.171954e+01  1.253983e+00 -8.244807e-01 -2.959612e-01
2.477226e+00
##  [881] -1.457223e+00  1.385689e+01  2.488584e+00 -1.046800e+01
3.673037e+00
##  [886] -1.168033e+01  1.687559e+01  1.318250e+00  1.146839e+01 -
1.003060e+01
##  [891]  3.772247e-01 -3.023802e+01  8.379725e-01  5.694992e-01  2.828060e-
01
##  [896]  2.329769e+00  3.886957e-01  8.327541e+00  2.366177e+00 -7.143309e-
01
##  [901]  4.554612e+01  1.131856e+01 -1.551958e+00 -2.688194e+00 -
9.381946e+00
##  [906]  1.034398e+01 -2.136868e+01  6.813014e-01  1.311847e+00 -3.314896e-
01
##  [911] -3.038105e+01  5.467337e+00  1.488270e+00  9.447085e+00 -
1.774411e+01
##  [916] -1.149409e+01 -1.556202e+01  2.612885e+01 -2.049403e+00 -
1.658487e+01
##  [921] -2.466374e+00  4.596561e+00  2.559549e+00  8.188882e-01 -
```

```
8.411714e+00
##  [926] -6.309726e-01  3.905097e+00 -1.256052e+01 -2.826341e+01 -
1.979030e+00
##  [931]  7.183143e-01  2.801257e+00  2.474780e+00 -4.271984e-01
2.746505e+00
##  [936]  5.756579e+00  1.145065e+01  2.830120e+00 -1.650013e+00 -
2.486818e+00
##  [941] -1.233633e+00  1.126428e+01 -1.508867e-01 -5.014240e-01 -3.830136e-
01
##  [946] -6.058968e+00  3.167819e+00 -2.168412e+01 -1.803550e-01
6.355494e+00
##  [951] -5.572872e-02  1.718233e+01 -3.949252e+00 -1.683928e+00
1.559402e+00
##  [956]  4.424014e+01 -5.963269e-01 -1.322388e+00  7.382759e+00 -
1.474951e+00
##  [961] -2.284346e+01 -5.010144e-01 -4.065636e-01  1.830258e+00
1.885188e+00
##  [966] -4.796761e-01  5.108027e+01  9.797859e-01  3.627773e-01
3.594442e+00
##  [971]  6.912145e-01  8.686127e+00  8.332253e+00 -1.023831e-02 -
1.880971e+00
##  [976]  6.553598e+00  5.717091e+01 -9.880275e-01  9.219671e+00  6.238708e-
01
##  [981] -2.352630e+01  5.670541e+00 -5.533830e-01 -4.317297e+00 -6.929923e-
01
##  [986]  2.302519e+01  4.615863e+00  2.266251e-01 -5.680107e-01 -
1.837568e+01
##  [991] -9.689318e+01  6.118889e+00 -1.259321e+01  4.112391e+00 -
9.247745e+00
##  [996]  2.631770e-01 -2.687153e+01 -1.452030e-01  5.282273e+00
6.311336e+00
```

```
data=data.frame(x1i,x2i,yi1)
train=data[1:800,]
train
```

```
##              x1i x2i           yi1
## 1   -1.120951293   0  1.258197e+01
## 2   -0.460354979   0 -7.998900e-01
## 3    3.117416628   0 -8.693620e-01
## 4    0.141016783   1  2.793949e+00
## 5    0.258575470   1  3.093778e+00
## 6    3.430129974   2 -1.560322e+01
## 7    0.921832412   1  4.086159e-01
## 8   -2.530122469   0 -7.979896e-01
## 9   -1.373705704   0  1.669925e+00
## 10  -0.891323940   0 -1.546481e-01
## 11   2.448163595   3 -3.276015e+01
## 12   0.719627654   1 -5.070414e-01
## 13   0.801542901   2 -1.466392e+00
```

```
## 14   0.221365432   2  2.471581e+00
## 15  -1.111682270   5  3.834323e+01
## 16   3.573826274   0  5.612161e-01
## 17   0.995700956   2 -3.772466e+00
## 18  -3.933234313   1  9.219498e+00
## 19   1.402711803   1  7.129518e-01
## 20  -0.945582815   0 -1.712949e+00
## 21  -2.135647412   6  1.144346e+02
## 22  -0.435949829   1  2.764449e+00
## 23  -2.052008897   6  1.092302e+02
## 24  -1.457782459   0  1.553209e-01
## 25  -1.250078536   0  1.271096e+00
## 26  -3.373386621   1  7.918368e+00
## 27   1.675574089   2 -5.398034e+00
## 28   0.306746236   2 -5.335542e-01
## 29  -2.276273874   1  5.961486e+00
## 30   2.507629842   2 -1.152669e+01
## 31   0.852928443   2 -7.933889e-01
## 32  -0.590142966   2  6.643167e+00
## 33   1.790251322   2 -7.719088e+00
## 34   1.756266975   0 -2.012163e-01
## 35   1.643162163   0  5.373037e-01
## 36   1.377280508   2 -9.294120e-02
## 37   1.107835307   2 -4.490548e+00
## 38  -0.123823421   4  7.096202e-01
## 39  -0.611925327   4  1.247489e+01
## 40  -0.760942002   2  7.961917e+00
## 41  -1.389413958   1  4.747887e+00
## 42  -0.415834556   0 -2.014455e-01
## 43  -2.530792703   1  6.424445e+00
## 44   4.337911931   2 -2.179528e+01
## 45   2.415923997   3 -3.191781e+01
## 46  -2.246217166   3  3.293195e+01
## 47  -0.805769671   3  1.114288e+01
## 48  -0.933310707   0 -6.672492e-01
## 49   1.559930237   0 -1.407981e+00
## 50  -0.166738133   1  2.535873e+00
## 51   0.506637028   2 -4.168187e-01
## 52  -0.057093511   0 -1.241326e+00
## 53  -0.085740915   4 -1.404207e+00
## 54   2.737204568   0  9.906046e-01
## 55  -0.451541971   3  7.345830e+00
## 56   3.032941209   0 -1.264157e+00
## 57  -3.097505608   2  2.188963e+01
## 58   1.169227499   2 -5.934670e+00
## 59   0.247708488   2  1.316911e-01
## 60   0.431883137   2  5.877678e-01
## 61   0.759278966   0  6.747910e-01
## 62  -1.004646906   3  1.487290e+01
## 63  -0.666414767   2  6.709744e+00
```

```
## 64  -2.037150766   3  2.685457e+01
## 65  -2.143582453   0 -4.034969e-01
## 66   0.607057283   0 -1.343382e+01
## 67   0.896419557   1  1.193136e+00
## 68   0.106008453   1  2.461323e+00
## 69   1.844534936   3 -2.432532e+01
## 70   4.100169371   3 -5.482320e+01
## 71  -0.982062332   0 -4.042175e-01
## 72  -4.618337751   3  6.210880e+01
## 73   2.011477049   2 -9.863387e+00
## 74  -1.418401525   1  2.785627e+00
## 75  -1.376017233   2  9.214869e+00
## 76   2.051142739   2 -8.541381e+00
## 77  -0.569546014   1  3.430865e+00
## 78  -2.441435425   1  6.585786e+00
## 79   0.362606959   4 -1.117888e+01
## 80  -0.277782725   0 -9.720281e-01
## 81   0.011528372   0 -1.327584e+00
## 82   0.770560802   2 -1.179995e+00
## 83  -0.741320064   2  5.690049e+00
## 84   1.288753097   0  6.567108e-01
## 85  -0.440973124   1  2.262513e+00
## 86   0.663563928   2 -1.080488e+00
## 87   2.193678026   1 -1.342175e+00
## 88   0.870362982   5 -3.622094e+01
## 89  -0.651863171   2  8.354025e+00
## 90   2.297615237   1 -9.777136e-02
## 91   1.987007712   0 -2.063142e+00
## 92   1.096793919   1  3.447791e+00
## 93   0.477463470   0 -1.943384e+00
## 94  -1.255812152   0  8.984240e-02
## 95   2.721304897   1 -1.215269e+00
## 96  -1.200519174   2  9.939187e+00
## 97   4.374665986   4 -1.076463e+02
## 98   3.065221252   1  2.055334e-02
## 99  -0.471400718   1  8.503111e-01
## 100 -2.052841801   0  9.743278e-01
## 101 -1.420813127   2  1.075482e+01
## 102  0.513767418   1  2.173421e+00
## 103 -0.493383757   1  2.676526e+00
## 104 -0.695085199   4  1.365856e+01
## 105 -1.903237135   2  1.408995e+01
## 106 -0.090055450   2  3.012768e+00
## 107 -1.569808939   0  2.817874e+00
## 108 -3.335883873   0  1.028763e+00
## 109 -0.760453041   3  1.227032e+01
## 110  1.837993218   3 -2.307137e+01
## 111 -1.150693925   2  1.066609e+01
## 112  1.215928644   1  2.407020e+00
## 113 -3.235765417   0  5.173503e-01
```

```
## 114 -0.111123931   1  1.980842e+00
## 115  1.038814408   1 -6.133606e+01
## 116  0.602306724   0 -1.114941e+01
## 117  0.211352388   2  1.397279e+00
## 118 -1.281412017   0  1.932550e+00
## 119 -1.699408692   2  1.321663e+01
## 120 -2.048257581   3  2.618815e+01
## 121  0.235293194   3 -3.394025e+00
## 122 -1.894949228   2  1.926320e+01
## 123 -0.981114887   1  4.502656e+00
## 124 -0.512184384   0  9.094044e-01
## 125  3.687724010   3 -4.965575e+01
## 126 -1.303899803   0  8.670288e-01
## 127  0.470773145   3 -5.587030e+00
## 128  0.155921699   1  3.819246e+00
## 129 -1.923713268   1  5.473459e+00
## 130 -0.142616172   3  3.115809e+00
## 131  2.889101717   0 -2.056721e+00
## 132  0.903008106   1 -2.830089e-01
## 133  0.082465844   0 -1.828252e+00
## 134 -0.844993665   0  2.143648e-01
## 135 -4.106494443   2  2.732115e+01
## 136  2.262674427   2 -1.105513e+01
## 137 -2.921280142   1  8.421853e+00
## 138  1.479895022   3 -1.915111e+01
## 139  3.818207138   4 -9.485269e+01
## 140 -2.887786322   1  7.432452e+00
## 141  1.403568671   2 -4.739031e+00
## 142 -0.524394979   0 -9.919621e-01
## 143 -3.144288318   1  8.992412e+00
## 144 -3.029335308   0  1.499215e+00
## 145 -3.203072347   0  1.151995e+00
## 146 -1.061813044   0 -5.180428e+00
## 147 -2.923511170   1  5.174534e+00
## 148  1.375833546   1 -2.757280e-01
## 149  4.200217881   2 -2.140987e+01
## 150 -2.574060952   2  1.809151e+01
## 151  1.575477695   1 -1.709098e+00
## 152  1.538084482   4 -3.973562e+01
## 153  0.664405158   2  4.221638e-02
## 154 -2.016753217   0  7.869057e-01
## 155 -0.238905213   1  1.935969e+00
## 156 -0.560790670   1  1.680271e+00
## 157  1.125979066   1  1.624263e+00
## 158 -0.744877512   3  1.041955e+01
## 159  1.953946773   1  6.743212e-01
## 160 -0.749161716   1  1.543682e+00
## 161  2.105422931   4 -5.155915e+01
## 162 -2.098354013   0  1.331510e-01
## 163 -2.520310490   2  1.864046e+01
```

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## 164  6.482079870   2 -3.771883e+01
## 165 -0.833715176   2  6.997015e+00
## 166  0.596455183   6 -3.358656e+01
## 167  1.273139348   0  1.266958e+00
## 168 -0.967561251   5  3.426574e+01
## 169  1.033724089   2 -3.325238e+00
## 170  0.737929055   1  2.602794e+00
## 171 -0.430761015   0 -9.996831e-01
## 172  0.130586067   1  3.196266e+00
## 173 -0.068134507   0 -5.162646e-01
## 174  4.256903798   1 -4.425589e+00
## 175 -1.482672193   3  2.119513e+01
## 176 -2.191992534   0  1.493901e+00
## 177  0.075576798   1  3.150113e+00
## 178  0.620961499   3 -8.582913e+00
## 179  0.873046958   0  2.282061e-01
## 180 -0.916730665   2  9.531491e+00
## 181 -2.126652268   4  4.860423e+01
## 182  2.526370352   2 -1.144247e+01
## 183 -0.699300776   0 -9.793750e-01
## 184 -1.731025725   1  7.066277e+00
## 185 -0.472559138   2  4.706802e+00
## 186 -0.394351789   0  1.494112e-01
## 187  2.219840579   3 -3.021594e+01
## 188  0.169474584   2 -8.059611e-01
## 189  1.508107570   1 -7.530767e-01
## 190 -0.998584034   2  8.771374e+00
## 191  0.428890619   2  9.635507e-01
## 192 -0.649371823   0 -2.344338e+00
## 193  0.189167056   2  2.860122e-01
## 194 -1.790726716   3  2.604434e+01
## 195 -2.621603067   3  3.554999e+01
## 196  3.994426769   2 -2.061965e+01
## 197  1.201417647   2 -5.667290e+00
## 198 -2.502542723   2  1.766957e+01
## 199 -1.222331833   1  4.023840e+00
## 200 -2.370960169   0  1.020128e-01
## 201  4.397620698   3 -5.910390e+01
## 202  2.624825953   2 -1.333916e+01
## 203 -0.530290113   2  2.812388e+00
## 204  1.086388118   0  2.208371e+00
## 205 -0.828679896   0  9.347666e-01
## 206 -0.952493789   2  7.263283e+00
## 207 -1.577205676   0  1.645729e-01
## 208 -1.189234535   2  1.111958e+01
## 209  3.301814935   2 -1.595934e+01
## 210 -0.108056250   1  3.835266e+00
## 211  0.238490473   1  9.352444e-01
## 212  0.487374859   2  2.300280e+00
## 213  2.464951757   0 -1.081025e+00
```

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## 214 -1.032127662   0 -2.057877e+00
## 215 -1.985014301   1  5.974405e+00
## 216  3.351393865   0 -3.753530e-01
## 217 -0.882326434   0 -1.402322e+00
## 218 -1.446131940   2  1.018774e+01
## 219 -2.472546238   2  1.941825e+01
## 220 -2.569431445   1  6.643557e+00
## 221 -1.147946959   1  4.170509e+00
## 222  1.235971634   2 -5.349353e+00
## 223  2.219696278   3 -2.977195e+01
## 224  1.415176708   0 -9.612843e-01
## 225 -0.727314594   5  2.448632e+01
## 226  0.119499875   0 -2.651897e+00
## 227 -1.409192927   0 -8.586633e-01
## 228 -1.434436323   1  2.124118e+00
## 229  1.769300998   3 -2.395705e+01
## 230 -2.031185157   0  2.476203e-01
## 231  3.910587931   2 -1.902276e+01
## 232 -0.180639188   3  1.260358e+00
## 233  0.429077653   1  2.110051e+00
## 234 -1.477055409   2  1.224068e+01
## 235 -1.148777380   1  2.922236e+00
## 236 -2.634032265   1  6.591319e+00
## 237 -0.365850777   4  7.178095e+00
## 238  0.837964810   1  3.769445e-01
## 239  0.648608688   0  7.588001e+00
## 240 -1.563072974   4  3.534211e+01
## 241 -1.577243942   1  7.376351e+00
## 242 -1.004397437   2  8.113516e+00
## 243  2.992121340   0 -1.030936e-01
## 244 -2.274607241   4  5.240414e+01
## 245 -0.358103189   3  6.044823e+00
## 246  3.804723643   3 -5.181423e+01
## 247 -0.201949771   1  1.456680e+00
## 248 -2.719681408   2  1.887269e+01
## 249 -1.329538871   1  5.182636e+00
## 250  0.970919958   1 -5.278234e-01
## 251 -0.751205743   7  5.649779e+01
## 252 -1.123752727   5  3.997284e+01
## 253 -0.687834468   1  3.268331e+00
## 254  0.180993294   2  4.662327e-01
## 255  3.197017542   1 -1.691894e+00
## 256 -0.177130224   1  4.188181e+00
## 257  2.161598992   2 -9.609654e+00
## 258  1.261508231   1  1.469724e+00
## 259 -0.227279791   0 -1.117415e+00
## 260 -3.065804006   5  1.126889e+02
## 261 -1.042234635   2  7.820138e+00
## 262 -0.979740906   3  1.189161e+01
## 263  0.094308866   1  2.198629e+00
```

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## 264  2.600397355   0  1.246410e-02
## 265  4.586157948   2 -2.231646e+01
## 266  3.095162118   0 -1.183142e+00
## 267 -0.266301929   0 -2.905262e-01
## 268 -3.513054791   3  4.871622e+01
## 269 -0.777559728   0  3.016673e-01
## 270  0.178414446   1  2.095357e+00
## 271  1.690026008   1 -5.044180e-01
## 272  1.925055937   1 -2.317115e-01
## 273  1.368618859   0  2.126738e+00
## 274 -2.790548700   0  5.427054e-01
## 275  1.699286091   3 -2.349801e+01
## 276 -0.893114433   4  2.042391e+01
## 277  0.349605400   4 -1.028265e+01
## 278  0.149102354   1  2.120216e+00
## 279  0.856333530   1  5.216008e-01
## 280  0.049349966   0 -6.695815e-01
## 281 -3.334950195   1  7.911052e+00
## 282  1.472991930   3 -2.064351e+01
## 283  0.772053137   1  1.168999e+00
## 284 -0.531303251   1  2.253174e+00
## 285  0.236289022   1  2.210803e+00
## 286  0.268077291   0  4.007634e-01
## 287  0.442038937   1  2.161838e+00
## 288  3.281692332   0  1.063313e+00
## 289 -0.438100758   2  4.218755e+00
## 290  0.336130768   1  1.896121e+00
## 291  2.336767746   1 -2.104084e-01
## 292  2.108362047   3 -2.892793e+01
## 293  2.290526221   0  1.626710e+00
## 294 -1.154936002   1  2.880383e+00
## 295  4.004965461   3 -5.344141e+01
## 296  0.133401742   1  2.331018e+00
## 297  3.733703689   1 -2.845967e+00
## 298 -2.701805372   3  3.683323e+01
## 299  0.041967173   2  2.876824e+00
## 300  2.499829142   3 -3.425680e+01
## 301 -1.430484374   1  1.968924e+00
## 302 -1.505377936   3  1.991809e+01
## 303 -1.877077407   0  1.215665e-01
## 304 -2.105026559   2  1.707565e+01
## 305 -0.874319066   2  9.491087e+00
## 306  0.662358346   0  5.329864e+00
## 307 -4.028420996   2  2.740016e+01
## 308  0.423960867   2 -1.329212e-01
## 309  2.473350093   3 -3.281701e+01
## 310  4.075148036   1 -4.888409e+00
## 311  2.602351984   1 -9.515090e-01
## 312  1.513549528   0 -3.876767e-01
## 313 -3.453460798   1  8.163189e+00
```

```
## 314 -1.203013416   2  9.477575e+00
## 315 -0.704092913   2  6.233573e+00
## 316  1.407047806   3 -1.925554e+01
## 317 -0.211342668   0 -8.285110e-01
## 318 -2.517297256   2  1.914786e+01
## 319  3.368871416   0  2.372748e-01
## 320  1.822782584   2 -7.484871e+00
## 321  0.474860545   2 -1.441226e+00
## 322  2.436217221   1  5.134676e-01
## 323 -2.677548574   1  6.864624e+00
## 324  1.321640596   2 -9.486262e+00
## 325 -1.045824753   1  4.140596e+00
## 326  1.367491044   1  1.793044e+00
## 327 -0.121643909   1  1.195816e+00
## 328  1.265921426   0  1.916366e+00
## 329  2.671035230   1 -1.505593e+00
## 330  0.014580181   1  3.489471e+00
## 331  2.035117274   0  1.500212e-01
## 332 -2.376868070   0 -2.419737e-01
## 333 -1.443208881   2  9.328712e+00
## 334  3.038435423   2 -1.646999e+01
## 335  0.754775946   2 -1.973830e+00
## 336 -4.104445641   0  8.521784e-01
## 337 -2.728074904   0 -1.818242e+00
## 338 -0.401562031   4  8.586807e+00
## 339  1.731558809   3 -2.078157e+01
## 340 -0.203766511   3  1.944651e+00
## 341  1.248374944   2 -5.271169e+00
## 342  1.918010756   0 -9.455617e-01
## 343  3.342109658   3 -4.476954e+01
## 344  0.112033467   3 -2.504105e-01
## 345 -0.103963812   0  3.953486e-01
## 346 -3.506474718   0 -8.143890e-01
## 347  0.198655188   2  1.093356e+00
## 348 -1.143700116   1  2.932636e+00
## 349 -1.948019166   1  6.142047e+00
## 350 -0.359812462   1  1.377711e+00
## 351  2.029886345   2 -1.185809e+01
## 352 -3.985496977   3  5.574408e+01
## 353 -0.854558574   1  5.892028e+00
## 354  0.233274567   2  8.152342e-01
## 355 -1.786415140   1  6.513596e+00
## 356  0.667805885   0  4.374121e+00
## 357  0.822859841   0  2.370701e+00
## 358 -0.066072319   1  3.048130e+00
## 359 -4.931796388   0  2.198895e-01
## 360  5.142916292   1 -6.892341e+00
## 361 -0.410598515   1  4.094008e+00
## 362  1.302386563   2 -7.531438e+00
## 363  0.547532982   1  4.639495e-02
```

```
## 364  2.049346470   3 -2.723997e+01
## 365  1.635318893   1  2.124642e-01
## 366 -0.419586342   1  4.083328e+00
## 367  0.756335544   1 -2.246313e-01
## 368 -1.890817662   2  1.552580e+01
## 369  1.713846022   2 -5.707133e+00
## 370 -0.922076678   2  9.543980e+00
## 371  4.833546708   3 -6.363674e+01
## 372 -3.302097791   1  9.267877e+00
## 373 -0.927974486   0 -1.043057e+00
## 374  1.650759726   1 -2.713240e-01
## 375  1.020265094   1 -4.906840e+00
## 376 -1.178962077   1  4.985212e+00
## 377 -1.993561484   3  2.882571e+01
## 378  0.288951409   1 -4.280016e-02
## 379 -0.028614826   0 -2.175170e-01
## 380 -3.580562475   4  8.429280e+01
## 381  0.069102134   1  2.227562e+00
## 382  0.380460631   0 -2.512404e+00
## 383  0.349452794   4 -1.007798e+01
## 384 -2.110034085   1  5.181181e+00
## 385  0.952266557   5 -3.826740e+01
## 386  2.757140274   0  9.493835e-01
## 387  0.912472806   2 -1.657122e+00
## 388 -2.271176941   2  1.510640e+01
## 389 -0.871290939   0  9.969215e-02
## 390  0.692207239   2 -2.297956e+00
## 391 -1.294091263   2  9.053687e+00
## 392 -4.315292670   0  1.336078e+00
## 393  1.768501640   1  1.246037e+00
## 394 -1.658955223   1  5.799012e+00
## 395 -1.147120542   3  1.655139e+01
## 396  3.007801218   5 -1.165608e+02
## 397 -1.548289859   0  2.081279e-01
## 398  1.691463080   1  1.425203e-01
## 399 -2.521365758   2  1.779604e+01
## 400 -0.709084806   3  1.063709e+01
## 401 -0.147112038   2  5.102641e+00
## 402 -2.337302849   2  1.657307e+01
## 403 -1.269496530   1  3.893152e+00
## 404 -0.057683106   3  2.751646e+00
## 405  1.341391937   2 -1.517511e+01
## 406 -3.301093087   1  5.000843e+00
## 407 -0.699508478   1  3.931440e+00
## 408  1.512812877   2 -6.501389e+00
## 409 -1.077618320   2  8.098763e+00
## 410  0.454583843   1  3.050263e+00
## 411  0.984457140   3 -1.058292e+01
## 412  0.535670031   2 -1.612982e+00
## 413  1.306515359   0  1.240258e+00
```

```
## 414 -0.245417322   0 -5.552072e-01
## 415 -0.827353027   3  1.163710e+01
## 416 -5.286297904   0  1.236195e+00
## 417 -0.185882037   1  2.890975e+00
## 418  0.860569393   1 -3.352486e-01
## 419  1.070797682   2 -4.497298e+00
## 420 -1.110556703   1  4.572648e+00
## 421  3.559005820   2 -2.084219e+01
## 422  0.572848839   3 -6.325023e+00
## 423  0.252631717   1  4.162315e-01
## 424  2.544533559   1 -1.129936e+00
## 425 -1.436932443   2  1.093987e+01
## 426 -0.900677248   1  4.967616e+00
## 427  4.794904960   0  2.028574e-01
## 428  0.022258374   3  1.365417e-01
## 429  3.267136843   1 -1.561677e+00
## 430 -2.877013290   2  1.933216e+01
## 431 -0.381033604   2  4.772371e+00
## 432  0.756847807   1  2.118742e+00
## 433  0.600077090   0 -7.965266e+00
## 434 -2.011272519   2  1.419393e+01
## 435  0.038518549   0  1.229627e-01
## 436 -2.154841306   1  4.634204e+00
## 437  1.425406650   0 -4.284664e-01
## 438  2.169550180   2 -8.882157e+00
## 439 -4.449975393   0 -2.230719e+00
## 440  2.471386925   2 -1.371694e+01
## 441 -2.482088994   2  1.915179e+01
## 442  0.909538538   1 -6.486019e-01
## 443  1.319805276   0  1.447822e+00
## 444 -0.399779656   0 -6.453033e-01
## 445 -1.290227914   0  1.406943e+00
## 446  0.330642042   3 -3.323781e+00
## 447  0.877637401   1 -2.381591e-01
## 448  1.766605640   0 -1.995913e-01
## 449 -4.104673968   1  8.376439e+00
## 450 -3.272758536   2  2.532333e+01
## 451  2.860804682   2 -1.417117e+01
## 452  2.093257694   1  3.207358e-02
## 453  0.870577898   0  3.650942e-01
## 454  1.430356814   2 -4.645803e+00
## 455  1.834349835   3 -2.255163e+01
## 456 -5.321845597   5  1.962744e+02
## 457  2.220554193   2 -1.038641e+01
## 458 -0.969975193   1  4.130839e+00
## 459  0.461233661   2 -5.487294e-01
## 460 -0.590315602   1  5.226642e+00
## 461  1.743929908   0 -1.320483e+00
## 462 -0.696944898   2  8.864011e+00
## 463  1.037007532   4 -2.771358e+01
```

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## 464 -0.781369957   0  6.014644e-01
## 465 -2.185574418   1  4.049224e+00
## 466  2.420021021   1 -1.986893e-03
## 467  1.481800023   0  1.031659e+00
## 468  3.448524478   0 -1.244419e+00
## 469  0.130307865   1  1.415854e+00
## 470  2.250005492   0  2.566907e-01
## 471  3.950838108   0  1.164328e+00
## 472 -0.562964230   0 -1.301029e+00
## 473 -2.645902226   1  6.436634e+00
## 474 -0.478703134   2  7.123607e+00
## 475 -0.428082480   1  1.909106e+00
## 476  0.303361009   1  2.845454e+00
## 477  3.424609955   2 -1.724843e+01
## 478 -0.652287785   0 -1.701296e+00
## 479  0.746009312   3 -1.044155e+01
## 480 -0.455368130   1  3.859195e+00
## 481  0.040901417   1  2.575800e+00
## 482  0.628115327   1  2.703363e+00
## 483  2.656429392   2 -1.366266e+01
## 484  0.242636755   1  2.855932e-01
## 485  1.425684640   3 -1.828512e+01
## 486  1.557720060   0 -4.700721e-01
## 487  1.829546542   0  2.177270e-01
## 488 -1.148789104   2  1.122984e+01
## 489  3.253762428   1 -2.489879e+00
## 490 -0.761913479   2  8.787956e+00
## 491 -0.211568335   2  2.855341e+00
## 492  2.808100535   1 -6.330326e-01
## 493  2.588167812   2 -1.240029e+01
## 494 -2.179983745   3  3.020601e+01
## 495 -1.746142001   2  1.076950e+01
## 496 -2.716158119   3  3.731058e+01
## 497  0.363694385   3 -4.329506e+00
## 498  0.329681736   1  2.129926e+00
## 499  0.728229375   2 -1.229097e+00
## 500  1.104315428   0 -1.360870e-01
## 501 -1.203785692   3  1.603967e+01
## 502 -1.987397182   2  1.565953e+01
## 503  2.053570112   2 -9.917681e+00
## 504  1.502122605   1  1.821734e+00
## 505 -3.018333075   2  2.116965e+01
## 506 -0.190294902   1  2.068712e+00
## 507 -1.791895646   2  9.244874e+00
## 508 -4.141502141   2  2.705137e+01
## 509  0.300240262   3 -1.679423e+00
## 510 -0.158423419   1  2.074862e+00
## 511 -0.194738535   0 -1.298393e+00
## 512  0.432305084   1  2.323430e+00
## 513  1.764930329   0  5.465082e-01
```

```
## 514  0.411195009   1  2.460430e+00
## 515 -1.232871686   2  9.716745e+00
## 516 -1.469598504   2  1.242509e+01
## 517 -0.263605586   1  1.621607e+00
## 518  0.620033973   2 -1.141937e+00
## 519 -2.079360705   1  5.222357e+00
## 520 -0.368617738   1  3.630471e+00
## 521  1.934534520   2 -8.501180e+00
## 522 -0.216560182   2  5.092154e+00
## 523 -1.396841335   2  1.297243e+01
## 524 -0.551890337   0 -3.886254e-01
## 525  2.229297091   1 -1.192683e+00
## 526  1.100087922   4 -2.970776e+01
## 527  2.473351600   1 -1.304273e+00
## 528  0.278195714   0 -7.910574e-01
## 529  0.820550193   1  2.702270e+00
## 530 -1.116913825   4  2.323408e+01
## 531  1.210741338   3 -1.726023e+01
## 532 -1.012667084   1  3.391057e+00
## 533 -2.841131009   2  2.053493e+01
## 534  0.255985931   1  2.798876e+00
## 535  3.891702435   3 -5.298317e+01
## 536  1.601828679   0  2.571869e-01
## 537  2.330506780   1 -8.607131e-01
## 538  0.717711446   2 -5.049434e-01
## 539 -1.217114357   3  1.781897e+01
## 540 -0.404481710   3  5.484601e+00
## 541 -0.546496214   3  7.758061e+00
## 542 -0.937399560   0 -6.560454e-01
## 543  1.408334568   2 -5.374628e+00
## 544 -2.394727005   2  1.704144e+01
## 545  1.732732264   2 -6.748419e+00
## 546  1.728304972   3 -2.263787e+01
## 547 -2.397244718   1  6.259326e+00
## 548  1.278983996   4 -3.323320e+01
## 549  4.860453305   1 -5.391632e+00
## 550 -1.114430964   0  2.311926e+01
## 551  1.689808483   2 -7.229750e+00
## 552 -1.564403691   0 -3.051361e-03
## 553  2.221422836   1  1.885199e-01
## 554  0.499649440   1  3.074157e+00
## 555  3.303830783   3 -4.413840e+01
## 556 -2.917941457   2  2.079973e+01
## 557 -0.102595772   1  2.479864e+00
## 558 -1.053850352   0 -3.325616e+00
## 559 -0.394529738   1  3.206127e+00
## 560 -1.259157485   1  4.167299e+00
## 561 -1.667687161   1  7.107444e+00
## 562  1.157444749   3 -1.640831e+01
## 563 -2.175161428   1  7.483758e+00
```

```
## 564  2.968061863   1 -1.454584e+00
## 565 -2.372413171   1  6.467038e+00
## 566  0.202158303   3 -2.958307e+00
## 567  1.065978574   0  1.948449e-01
## 568  1.173470678   1  2.187051e+00
## 569 -0.603493328   0 -2.622780e+00
## 570  0.159003991   3 -2.415822e+00
## 571  1.922528304   1  4.242631e-01
## 572 -2.912931834   1  6.746255e+00
## 573 -1.563479423   1  7.485633e+00
## 574  0.640804629   1  6.135765e-01
## 575 -0.889563957   0 -8.162737e-01
## 576  2.740007988   0 -1.592474e+00
## 577  1.346507727   0  1.065666e+00
## 578  0.144333505   0 -2.744468e+00
## 579 -3.015514635   1  6.685476e+00
## 580  0.052200451   2  1.577968e+00
## 581 -0.632831736   2  5.222799e+00
## 582 -0.204693027   0 -1.539788e+00
## 583 -2.363118454   4  5.667565e+01
## 584  0.997316088   0  3.186885e-01
## 585 -2.077912881   1  5.624085e+00
## 586 -0.452443963   2  3.074596e+00
## 587  0.762851659   3 -1.132730e+01
## 588 -1.567031578   3  2.125586e+01
## 589  1.165982811   1  1.007221e+00
## 590 -2.633020803   1  5.845933e+00
## 591 -5.619549358   1  1.210006e+01
## 592  0.929935982   2 -2.880171e+00
## 593  1.681079654   2 -8.012903e+00
## 594 -0.571690847   0 -1.008490e+00
## 595  1.008252509   0  1.388944e-01
## 596 -2.311833050   3  3.137298e+01
## 597 -0.254297214   1  1.816242e+00
## 598 -3.883036752   2  2.609740e+01
## 599  2.362361783   3 -3.061003e+01
## 600  3.719821723   1 -2.719772e+00
## 601  2.148024521   1 -7.287907e-01
## 602 -0.054693933   2  3.397214e+00
## 603 -0.066660685   1  3.288814e+00
## 604 -3.032135246   3  4.131004e+01
## 605  1.580770688   0  5.898588e-01
## 606 -0.421468361   0 -1.627521e+00
## 607 -1.313485851   2  1.020230e+01
## 608 -2.824051585   0 -3.620728e-01
## 609 -0.599524993   2  6.468718e+00
## 610 -1.698122270   1  6.235020e+00
## 611 -0.794061041   2  7.305324e+00
## 612 -2.435199976   1  5.603980e+00
## 613  3.375178966   0 -9.017512e-01
```

```
## 614 -0.032005054   1  2.719656e+00
## 615  2.149890155   1  8.252090e-01
## 616 -5.203399341   2  3.326770e+01
## 617 -0.906395660   0 -1.512894e+00
## 618 -1.350964583   2  1.099203e+01
## 619 -2.445852359   1  5.829966e+00
## 620  3.093218308   4 -7.641094e+01
## 621 -2.830563840   3  3.870806e+01
## 622  0.636780521   0  1.405788e+01
## 623  1.692872571   1  1.065142e+00
## 624  0.356380377   2  1.368938e+00
## 625 -1.750510955   2  1.303760e+01
## 626  1.882331613   1  1.173389e+00
## 627  0.341176164   1  1.326219e+00
## 628 -2.126995813   0 -3.416639e-01
## 629 -2.776098095   1  7.106060e+00
## 630  4.173434869   0  6.484790e-01
## 631 -1.357006291   2  1.035524e+01
## 632 -3.711143308   2  2.469864e+01
## 633  1.066518726   1  5.605184e+00
## 634  0.620460512   1  2.777653e+00
## 635 -2.707668671   0 -2.443527e+00
## 636 -3.885912827   0 -1.400682e-01
## 637 -0.232605034   3  3.515539e+00
## 638  2.278792583   2 -1.182304e+01
## 639  1.272248070   2 -5.808801e+00
## 640 -0.985874848   1  1.803986e+00
## 641 -1.668376463   1  6.946093e+00
## 642  0.542133514   4 -1.711334e+01
## 643  0.314706694   0 -2.988779e+00
## 644  1.259423493   0 -5.343648e-01
## 645 -0.791595901   4  1.672876e+01
## 646  1.798708106   2 -8.412260e+00
## 647 -1.661623055   7  1.235378e+02
## 648 -0.661089400   1  3.698553e+00
## 649  1.481629047   2 -6.323682e+00
## 650  1.979943217   0 -3.480130e-03
## 651 -3.877009403   2  2.661111e+01
## 652  0.214380814   3 -9.294418e-01
## 653  1.217558023   3 -1.578352e+01
## 654 -2.901648627   0  1.769246e+00
## 655  0.961251200   1 -1.203855e-01
## 656 -1.656348541   2  1.211378e+01
## 657  2.040506024   1 -2.611941e+00
## 658  1.076964064   2 -5.297049e+00
## 659  1.538104590   0 -1.763920e-01
## 660  0.241438659   2  1.933049e+00
## 661  1.727296868   1 -1.503518e-01
## 662  2.761029057   1 -9.425468e-01
## 663  3.932496031   2 -2.181679e+01
```

```
## 664 -0.056790109   0 -7.571857e-01
## 665 -4.498102181   1  9.593649e+00
## 666  0.063052009   2  1.940283e+00
## 667  0.411122417   1  1.821764e+00
## 668 -0.310690694   1  2.193545e+00
## 669  1.136577242   1  2.634062e+00
## 670  2.021355929   0 -2.184006e-03
## 671 -1.035964860   0 -3.005598e+00
## 672 -0.588190654   0 -2.498190e+00
## 673  0.795684418   2 -2.864247e+00
## 674 -1.100447483   0 -3.624692e+01
## 675  0.182534767   0 -2.212578e+00
## 676 -3.923415198   0 -4.669634e-01
## 677 -2.239799443   1  5.954245e+00
## 678 -2.655510951   3  3.671472e+01
## 679 -1.707247393   1  6.944471e+00
## 680 -1.386609056   1  4.097106e+00
## 681  0.764610285   1  1.928495e+00
## 682  1.964226005   1 -7.570074e-01
## 683 -1.454767058   3  1.959353e+01
## 684 -1.993677965   3  2.749777e+01
## 685 -2.083377718   1  5.932049e+00
## 686 -0.829177453   2  5.956582e+00
## 687 -0.478058136   0 -1.344932e+00
## 688  0.967235068   4 -2.467215e+01
## 689 -0.642649684   0 -1.198277e+00
## 690 -4.156978538   5  1.532816e+02
## 691 -0.182868551   0 -4.489467e-01
## 692  2.374373621   1 -1.383016e+00
## 693  2.383202537   0 -2.946302e-01
## 694 -1.577926436   2  1.222117e+01
## 695 -3.095553087   2  2.166324e+01
## 696  4.916120984   6 -2.650852e+02
## 697 -0.324843885   0 -6.652038e-01
## 698 -0.194902500   3  2.045073e+00
## 699  0.841148377   2 -2.314019e+00
## 700 -3.228078913   5  1.188791e+02
## 701 -1.456438222   1  5.023893e+00
## 702 -3.080884811   2  2.090290e+01
## 703 -1.386189228   0  2.864874e-01
## 704  0.237698866   2  2.125943e+00
## 705 -2.729418916   0  1.344645e+00
## 706  1.179965359   3 -1.639403e+01
## 707  0.578688058   3 -6.834955e+00
## 708 -1.808430051   3  2.393495e+01
## 709  0.452649885   3 -6.752137e+00
## 710  1.496162325   1  1.238538e+00
## 711  2.122190507   4 -5.289681e+01
## 712 -0.425696557   1  3.422323e+00
## 713 -0.187273588   1  2.098904e+00
```

```
## 714 -0.173428270   1  4.453401e+00
## 715  2.882923512   0  7.035454e-01
## 716  2.250143785   3 -3.018939e+01
## 717  1.668803136   0  8.216190e-01
## 718 -0.574681600   1  2.031187e+00
## 719  0.746482868   2 -2.018141e+00
## 720  0.806580662   2 -8.563025e-01
## 721 -2.083346589   4  4.709905e+01
## 722 -3.456609031   0  9.464482e-01
## 723  1.283660055   1  9.672989e-01
## 724 -3.058621062   3  3.950747e+01
## 725  0.003367377   3 -5.902904e-01
## 726  0.500495643   3 -6.215917e+00
## 727  1.127734781   2 -3.018366e+00
## 728  0.378852475   0 -1.542734e+00
## 729 -1.465707613   0  7.350998e-01
## 730  1.972731720   0 -8.818157e-01
## 731  3.477267534   1 -3.595495e+00
## 732  1.762357618   2 -8.298691e+00
## 733 -3.887301803   3  5.432010e+01
## 734  2.799152371   1 -6.044885e-01
## 735 -0.112111892   3  2.443846e+00
## 736  1.049828559   1  1.730861e+01
## 737  1.244066472   2 -6.100958e+00
## 738 -0.193372146   2  4.602804e+00
## 739 -0.150526395   2  3.795463e+00
## 740  2.038314139   3 -2.793288e+01
## 741  1.423203845   1 -5.116631e-02
## 742  1.980524492   0  1.323716e-01
## 743  4.765853390   0  2.198062e-01
## 744  1.328831727   0 -1.206022e-01
## 745  0.414762315   3 -6.272983e+00
## 746 -4.421266221   2  2.727862e+01
## 747  5.383428006   0  4.363984e-01
## 748 -0.965353644   0 -1.836192e+00
## 749  4.749469430   0  5.634332e-01
## 750  0.749287136   1 -5.813245e-01
## 751  3.076860398   2 -1.685496e+01
## 752 -0.219420642   0 -5.726786e-01
## 753  1.022941509   0 -1.007414e+00
## 754  0.427915959   1 -3.044611e-01
## 755 -0.372241399   1  3.738622e+00
## 756 -0.240787649   1  3.660922e+00
## 757  2.025668672   0 -9.514436e-01
## 758 -0.402916294   3  5.503706e+00
## 759 -4.075364989   2  2.616540e+01
## 760 -0.391778498   1  3.300041e+00
## 761  1.079581211   2 -4.121497e+00
## 762  1.232911433   1  5.415036e-01
## 763  1.233135635   2 -6.294234e+00
```

```
## 764 -3.384203042   1  6.664668e+00
## 765  0.737484116   2 -1.387561e+00
## 766  1.935718420   1  3.873955e-01
## 767  2.553157363   2 -1.205820e+01
## 768 -0.449922542   1  2.225169e+00
## 769 -0.643785172   2  4.750423e+00
## 770  2.975675664   2 -1.446020e+01
## 771 -3.335856092   0 -1.379977e-01
## 772 -0.873659954   1  3.602619e+00
## 773  0.914924159   1  1.146667e+00
## 774 -3.235547531   1  6.615469e+00
## 775  0.559255725   1  2.570051e+00
## 776  3.755728042   3 -5.065055e+01
## 777 -0.008121307   2  2.015688e+00
## 778 -0.556908050   0 -7.375914e-01
## 779  0.949823428   0  9.038864e-01
## 780 -0.558144342   1  3.015246e+00
## 781  1.626800748   1 -5.453132e-01
## 782  1.808870929   2 -8.108613e+00
## 783  0.005383322   0 -3.043844e+00
## 784 -2.353384315   4  5.635385e+01
## 785 -2.636441455   3  3.663540e+01
## 786 -1.185994731   2  9.556383e+00
## 787  1.594761001   3 -2.561526e+01
## 788 -3.916410349   0  5.130109e-01
## 789 -3.772650317   2  2.613510e+01
## 790 -1.307559650   3  1.859570e+01
## 791  0.788789696   0 -6.380521e-01
## 792 -1.827132096   1  5.749707e+00
## 793  1.773498074   2 -5.965046e+00
## 794  0.666739940   1  7.073359e-01
## 795 -0.341279236   3  5.396941e+00
## 796  1.637656273   1  4.782753e-01
## 797  0.776730326   2 -1.513749e+00
## 798 -0.891870054   2  7.735195e+00
## 799  0.462229869   1  9.669107e-01
## 800  1.295026716   3 -1.729359e+01
```

```
test=data[-(1:800),]
test
```

```
##               x1i x2i          yi1
## 801   0.712566690   0   2.33514807
## 802  -1.316020416   1   3.69297865
## 803   1.710404410   1   0.23366864
## 804   2.305872451   0   0.47142044
## 805   0.552549113   2  -1.45312838
## 806   0.288209325   2   2.07670285
## 807  -0.151250162   1   4.82546553
## 808   4.322831693   3 -56.42372649
```

```
## 809   0.552631064   2  -1.13186634
## 810  -0.316588057   4   3.34947336
## 811  -5.015835604   3  68.22326395
## 812  -3.130563530   1   8.23612299
## 813  -0.155346401   2   4.39290643
## 814   0.412588090   3  -6.47677839
## 815   0.553744918   3  -8.27415065
## 816   1.643013561   0   0.25676477
## 817  -0.388304819   0   0.15132935
## 818   2.429177589   5 -93.01251918
## 819  -1.843032086   1   6.56397868
## 820  -2.416885441   2  16.50948508
## 821  -2.457972355   1   5.08840366
## 822   1.484594048   1  -1.37245875
## 823  -0.165839885   2   4.18278434
## 824   1.579635844   3 -22.89809849
## 825  -0.535412843   0  -0.18545944
## 826  -1.183784204   1   2.86919815
## 827  -0.736705150   2   8.65656218
## 828  -3.705233648   0  -0.33949629
## 829  -2.339230521   2  18.81863896
## 830  -2.884069293   1   7.55064844
## 831   2.108644544   2  -8.52708233
## 832  -1.194660183   2  10.10612994
## 833   1.578919706   2  -6.27294999
## 834   3.032981203   3 -41.47298098
## 835  -0.383549618   1   2.41307182
## 836   0.567757813   1   1.36142316
## 837  -3.502135039   3  48.04629543
## 838  -1.637339555   1   6.16477849
## 839   0.112429699   2   1.82353584
## 840   0.598173807   3  -7.58057337
## 841  -1.518796235   1 -22.83893751
## 842   5.369717998   1  -5.97039849
## 843  -0.916780278   3  11.24264176
## 844   0.128487113   2   2.75074493
## 845   1.299583734   2  -7.00333840
## 846  -0.052037270   2   2.90977515
## 847  -1.287134774   2   9.56247572
## 848   2.090611328   1  -0.68423623
## 849   3.231090646   1  -3.29174550
## 850  -0.059387942   0  -1.53170979
## 851   1.124534691   4 -31.39356861
## 852  -0.194824997   1   3.67253052
## 853   2.032910435   1  -0.21709914
## 854  -2.312334789   0  -1.01939324
## 855   4.641720448   2 -23.85152774
## 856  -1.207062496   1   4.55938731
## 857  -2.917698829   1   7.78997954
## 858  -0.701835654   3   9.57957788
```

```
## 859   0.293416968   1   1.49172618
## 860   3.247242415   2 -16.72687572
## 861   1.822419354   2  -7.51031721
## 862   0.284916864   0   0.06769007
## 863  -2.778967048   0   1.65005629
## 864  -1.732075489   1   5.29449584
## 865  -0.326569864   2   3.62658351
## 866   5.106052227   2 -27.70339756
## 867  -3.720455148   2  28.19971582
## 868   2.262109304   3 -30.40776709
## 869  -1.054468514   0  -2.82675388
## 870   3.331981800   1  -2.60151355
## 871  -2.278401280   2  16.22507336
## 872   0.287246469   0  -1.44648281
## 873  -2.199101883   0  -0.09946032
## 874   1.807032853   0   1.74179035
## 875   2.967558977   2 -14.64022716
## 876   3.901442013   3 -51.71954270
## 877   1.595201315   1   1.25398251
## 878   3.686532509   1  -0.82448072
## 879   2.492847817   1  -0.29596122
## 880  -0.263749824   4   2.47722645
## 881   0.954074477   1  -1.45722295
## 882  -1.943988423   2  13.85689274
## 883  -0.370404339   3   2.48858437
## 884   2.441927417   2 -10.46799712
## 885   1.082568287   1   3.67303717
## 886   0.914714663   3 -11.68032907
## 887  -2.076262079   2  16.87559434
## 888  -1.209026461   0   1.31824997
## 889  -1.529212028   2  11.46839126
## 890   0.790591740   3 -10.03059710
## 891  -1.981015261   0   0.37722466
## 892   1.124082780   4 -30.23801641
## 893  -2.232832824   0   0.83797255
## 894   3.657060916   0   0.56949917
## 895   0.921182699   0   0.28280603
## 896  -1.402007225   1   2.32976938
## 897   0.482091865   2   0.38869566
## 898  -0.704906398   2   8.32754142
## 899   0.742295920   1   2.36617722
## 900   0.487065438   0  -0.71433089
## 901  -2.028228345   4  45.54612078
## 902  -1.582627758   2  11.31856154
## 903   0.599187369   2  -1.55195816
## 904   3.278103818   1  -2.68819354
## 905   2.169234017   2  -9.38194570
## 906  -1.249134947   2  10.34397889
## 907   1.651845803   3 -21.36867672
## 908  -0.097136706   1   0.68130141
```

```
## 909   0.602627303   1   1.31184743
## 910   0.520722981   2  -0.33148957
## 911   5.150899527   2 -30.38105460
## 912  -2.370577621   1   5.46733746
## 913   0.201839718   2   1.48827048
## 914  -3.559954576   1   9.44708508
## 915   1.179671846   3 -17.74411323
## 916   2.193216943   2 -11.49409181
## 917   2.891324482   2 -15.56201847
## 918  -3.850290504   2  26.12885442
## 919   0.825538994   2  -2.04940265
## 920   3.186739903   2 -16.58487004
## 921  -0.828031726   0  -2.46637371
## 922  -0.424301064   2   4.59656092
## 923  -0.073074444   2   2.55954860
## 924   0.730037503   1   0.81888823
## 925   1.330319751   2  -8.41171354
## 926   2.635641767   0  -0.63097263
## 927  -0.190975181   2   3.90509744
## 928   0.392556091   4 -12.56052440
## 929   4.975995754   2 -28.26340840
## 930   0.862197856   2  -1.97903029
## 931   0.377506219   1   0.71831428
## 932  -2.684486251   0   2.80125733
## 933   0.005712095   1   2.47478047
## 934  -0.442652306   0  -0.42719840
## 935  -0.022091660   2   2.74650487
## 936  -1.150835283   1   5.75657919
## 937  -1.373631305   2  11.45065300
## 938  -1.441547264   1   2.83012020
## 939  -0.429009031   0  -1.65001287
## 940   2.736265296   1  -2.48681814
## 941   2.098173253   1  -1.23363314
## 942  -0.719950236   3  11.26427698
## 943  -3.371832909   0  -0.15088674
## 944  -1.689166857   0  -0.50142400
## 945  -0.915521066   0  -0.38301359
## 946   0.207276008   4  -6.05896802
## 947  -1.325214551   1   3.16781866
## 948   4.013361382   2 -21.68412021
## 949  -0.544535069   0  -0.18035502
## 950  -2.427888940   1   6.35549441
## 951  -0.282523515   0  -0.05572872
## 952  -2.010755163   2  17.18233381
## 953   0.312311415   3  -3.94925197
## 954   0.467267228   0  -1.68392791
## 955   0.711175223   1   1.55940193
## 956  -3.243716518   3  44.24013684
## 957   0.441422582   0  -0.59632690
## 958   0.620900161   2  -1.32238777
```

```
## 959  -2.842216895   1   7.38275876
## 960   1.910731279   0  -1.47495139
## 961   1.568341758   3 -22.84346331
## 962   4.599238722   0  -0.50101440
## 963   0.313405975   2  -0.40656355
## 964   0.093467055   1   1.83025821
## 965   0.193171668   2   1.88518831
## 966   0.139532462   0  -0.47967607
## 967  -3.696945551   3  51.08027263
## 968  -3.342254117   0   0.97978589
## 969  -0.155077934   0   0.36277734
## 970  -1.162134761   1   3.59444243
## 971   0.109473050   2   0.69121452
## 972  -4.222416747   1   8.68612668
## 973  -2.997396509   1   8.33225317
## 974  -2.202966878   0  -0.01023831
## 975   1.972116442   1  -1.88097146
## 976  -2.196980014   1   6.55359811
## 977  -1.599027907   5  57.17090550
## 978   0.159747637   0  -0.98802755
## 979  -0.645492724   3   9.21967085
## 980   0.292834357   0   0.62387077
## 981   4.610123964   2 -23.52630223
## 982  -2.249207343   1   5.67054073
## 983  -0.610939280   0  -0.55338298
## 984  -1.033518899   0  -4.31729673
## 985   3.024790854   0  -0.69299230
## 986  -1.538969846   3  23.02519166
## 987  -0.164173808   5   4.61586293
## 988   1.574267229   1   0.22662507
## 989  -2.117181072   0  -0.56801069
## 990   3.310351631   2 -18.37568082
## 991   1.351524831   2 -96.89317854
## 992  -2.148413219   1   6.11888911
## 993   0.909155618   3 -12.59320596
## 994  -0.426614286   1   4.11239133
## 995   0.626457544   3  -9.24774516
## 996  -0.179950394   0   0.26317696
## 997   2.141032074   3 -26.87153353
## 998  -2.702200771   0  -0.14520302
## 999  -1.045233394   1   5.28227266
## 1000 -0.498381356   2   6.31133576
```

```
tr_k_1=knnreg(yi1~x1i+x2i,data=train,k=1)
Y_KNN=predict(tr_k_1,test)
mse_knn_1=mean((test$yi1-Y_KNN)^2)
mse_knn_1
```

```
## [1] 47.5249
```

We see knn value is coming to e higher and lower for simple models.